

ANNEXURE – ‘A’

ENCLOSED TECHNICAL DOCUMENTS & DRAWINGS

TABLE II**Indicative list of existing Bag Filter Equipments/components**

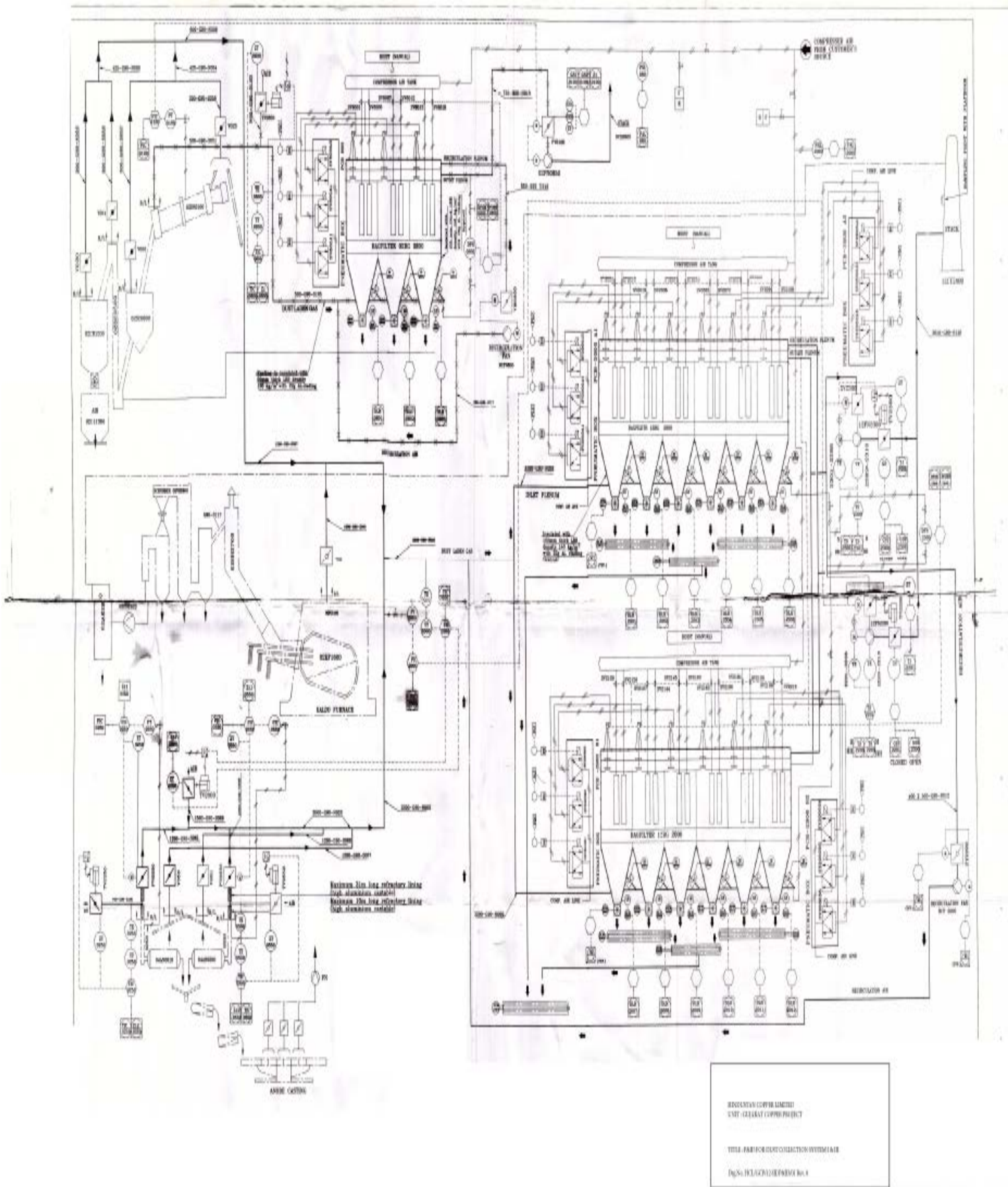
Sr. No	Item description	Quantity	Remarks
1	Casing with inlet & outlet plenum	Lot	
2	Nozzle house, Pressure Tank, bag holding plate etc.	Lot	
3	Hopper	12 nos.	
4	Slide Gate	12 nos.	
5	Support Structure	Lot	
6	Platforms & Stairs	Lot	All installed at various levels
7	Canopy Structure	Lot	All installed
8	GI sheet for canopy structure	Lot	All installed
9	Hot Insulation for Bag Filter casing (top, sides & Hopper, re-circulation duct & Inlet /outlet duct along with Al cladding	Lot	As installed
10	Bag cage	App.4000 nos	Including spare cages lying on floor & cages have fallen into the hoppers
11	Polyester Bags	App. 3500 nos	
12	Hopper heating jacket	Lot	
13	Hopper Heaters	48 nos	Along with Control Panels
14	Thermostat		Refer list under Instrumentation
15	Rotary Feeder	12 nos	Along with Geared motor
16	Pneumatic Cylinders for outlet & re-circulation Dampers	24 nos	
17	Screw Conveyors	7 nos	Along with drives
18	Pneumatic pipelines & fittings	Lot	From Air receiver to Bag-filter
19	Solenoid valves	App. 220 nos	Free Instrumentation List
20	Inlet & outlet Dampers	Lot	
21	Pressure switch	Lot	
22	Dust Level Indicator	12nos	
23	Pressure Gauges	12 nos	
24	Filter Regulator Unit	Lot	
25	DP Switch	Lot	
26	Manual Chain Pulley Block		Along with mono-rail
27	Flexible connections for Rotary Feeders & Screw Conveyors	Lot	
28	Re-circulation Fan	One	Along with motor ,base frame etc
29	Inlet Butterfly Damper for re-circulation Fan	One	
30	Duct for re-circulation Fan	Lot	
31	Motorized actuator for Re-circulation Fan and other Dampers		Details given in Table III

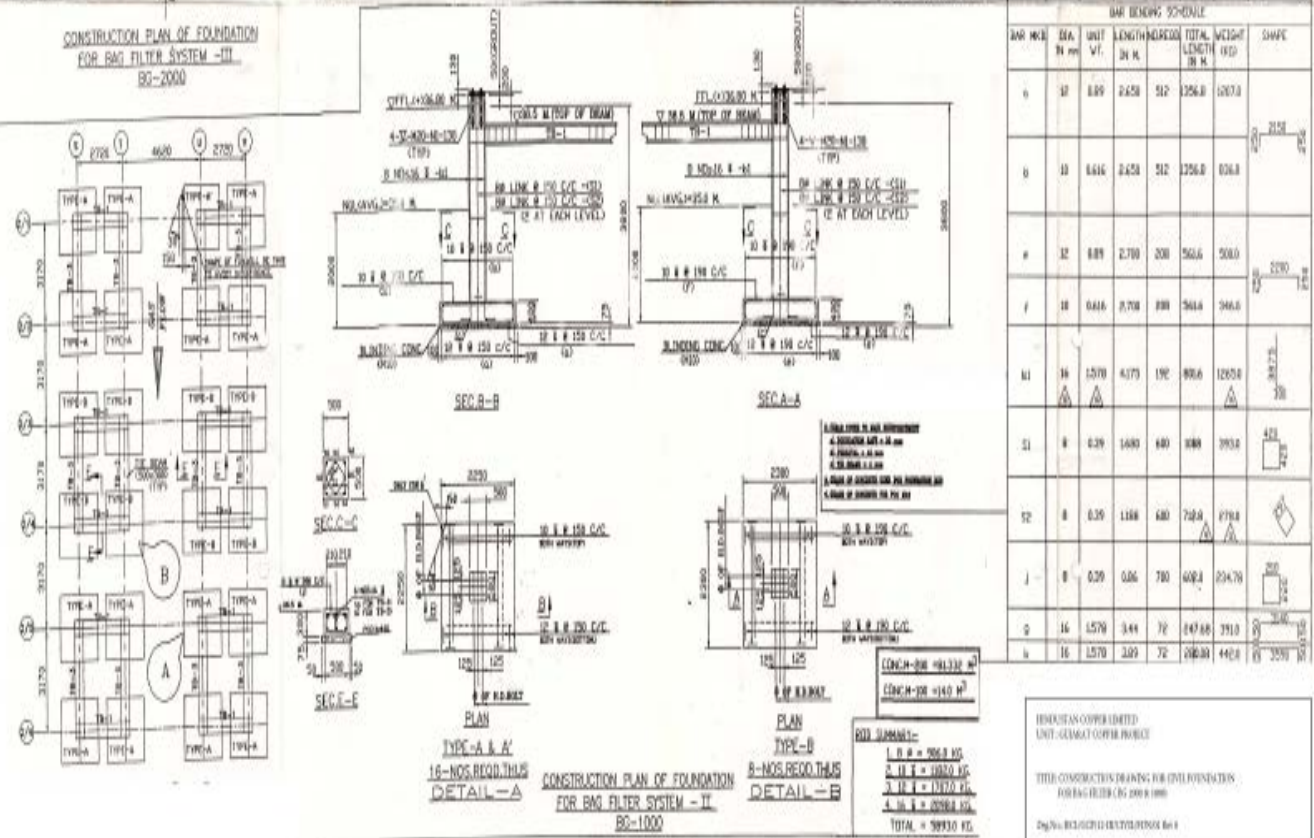
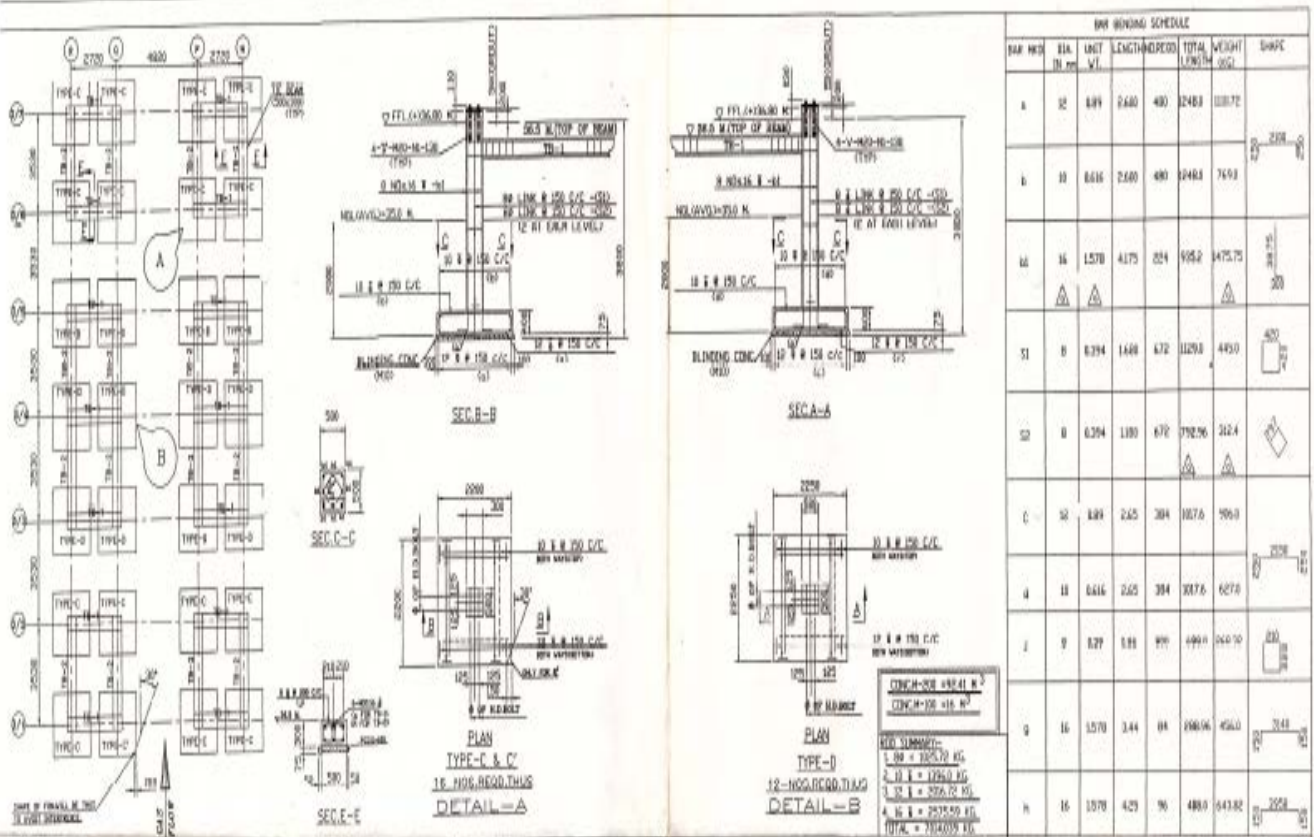
TABLE III
LIST FOR DAMPERS

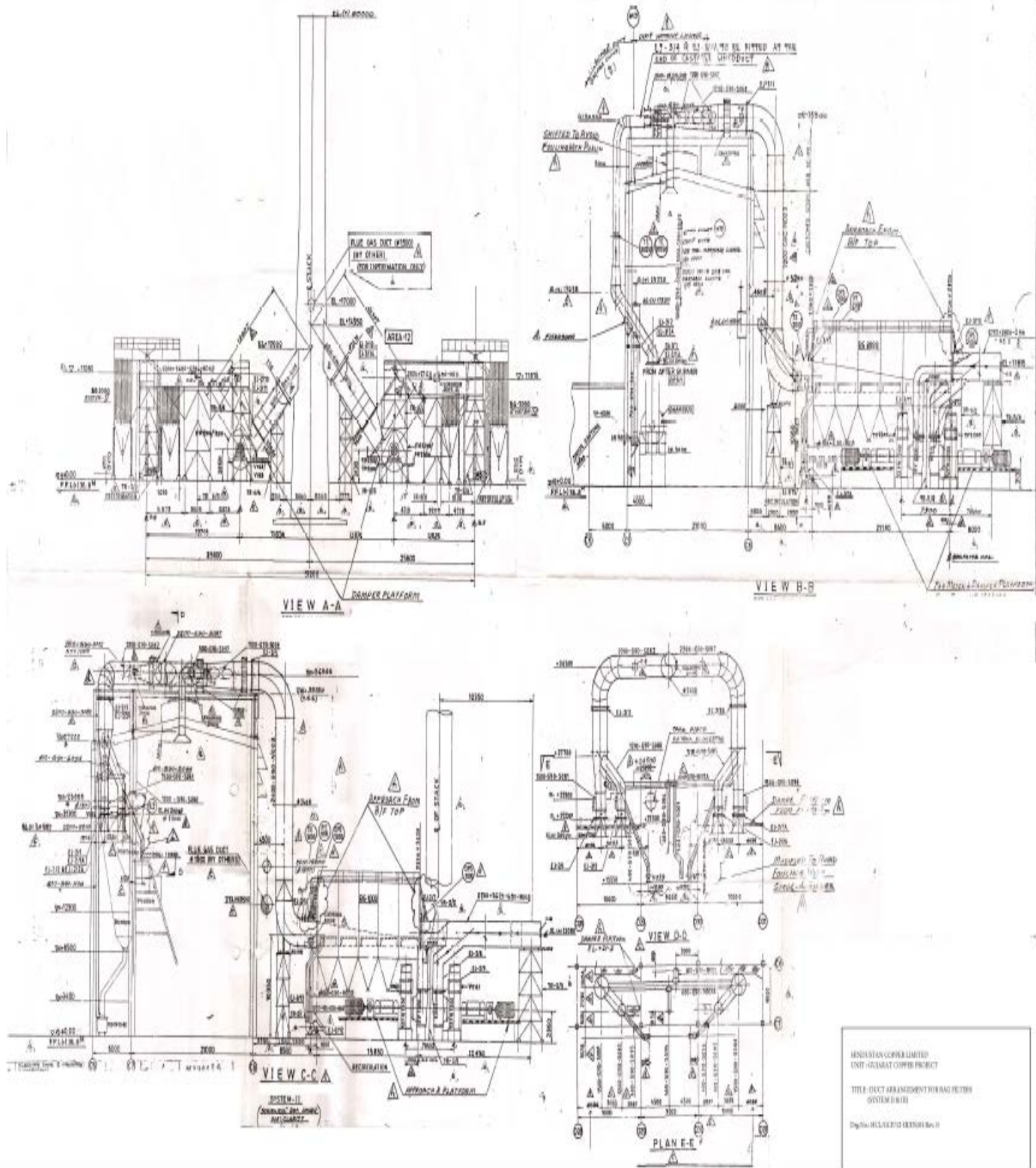
Sr. No.	Tag No	Type	Line Size	Line No	Location	Remarks
1	TV 2000	Pneumatic	1500	1500-G90-S069	Inlet Gas duct	
2	TV 0250	Pneumatic	700	700-G90-S100	Inlet Line to After Burner for AN 0010	
3	TV 0550	Pneumatic	700	-	Inlet Line to After Burner for AN 0300	
4	PV 0250	Motorized	1200	1200-G90-S065	After Burner Outlet Duct	
5	PV 0550	Motorized	1200	1200-G90-S066	After Burner Outlet Duct	
6	V 0050	Manual	1200	1200-G90-S067	Ventilation at Shed Roof	
7	V 0051	Manual	1200	1200-G90-S068	Ventilation at Shed Roof	
8	ID 2001	Manual	3200	3200-G90-N003	Hopper Inlet Damper	
9	ID 2002	Manual	3200	3200-G90-N003	Hopper Inlet Damper	
10	ID 2003	Manual	3200	3200-G90-N003	Hopper Inlet Damper	
11	ID 2004	Manual	3200	3200-G90-N003	Hopper Inlet Damper	
12	ID 2005	Manual	3200	3200-G90-N003	Hopper Inlet Damper	
13	ID 2006	Manual	3200	3200-G90-N003	Hopper Inlet Damper	
14	ID 2007	Manual	3200	3200-G90-N003	Hopper Inlet Damper	
15	ID 2008	Manual	3200	3200-G90-N003	Hopper Inlet Damper	
16	ID 2009	Manual	3200	3200-G90-N003	Hopper Inlet Damper	
17	ID 2010	Manual	3200	3200-G90-N003	Hopper Inlet Damper	
18	ID 2011	Manual	3200	3200-G90-N003	Hopper Inlet Damper	
19	ID 2012	Manual	3200	3200-G90-N003	Hopper Inlet Damper	
20	XV 2300	Motorized	-	-	12 FN 2300 Inlet Line	
21	TV 2300	Pneumatic	-	-	12 FN 2300 Discharge Line	
22	XV 2200	Motorized	-	-	12 FN 2200 Inlet Line	
23	TV 2200	Pneumatic	-	-	12 FN 2200 Discharge Line	
24	PV 2000	Motorized	400x500	400x500-G90-N012	Recirculation Air Fan RCF 2000 Inlet	

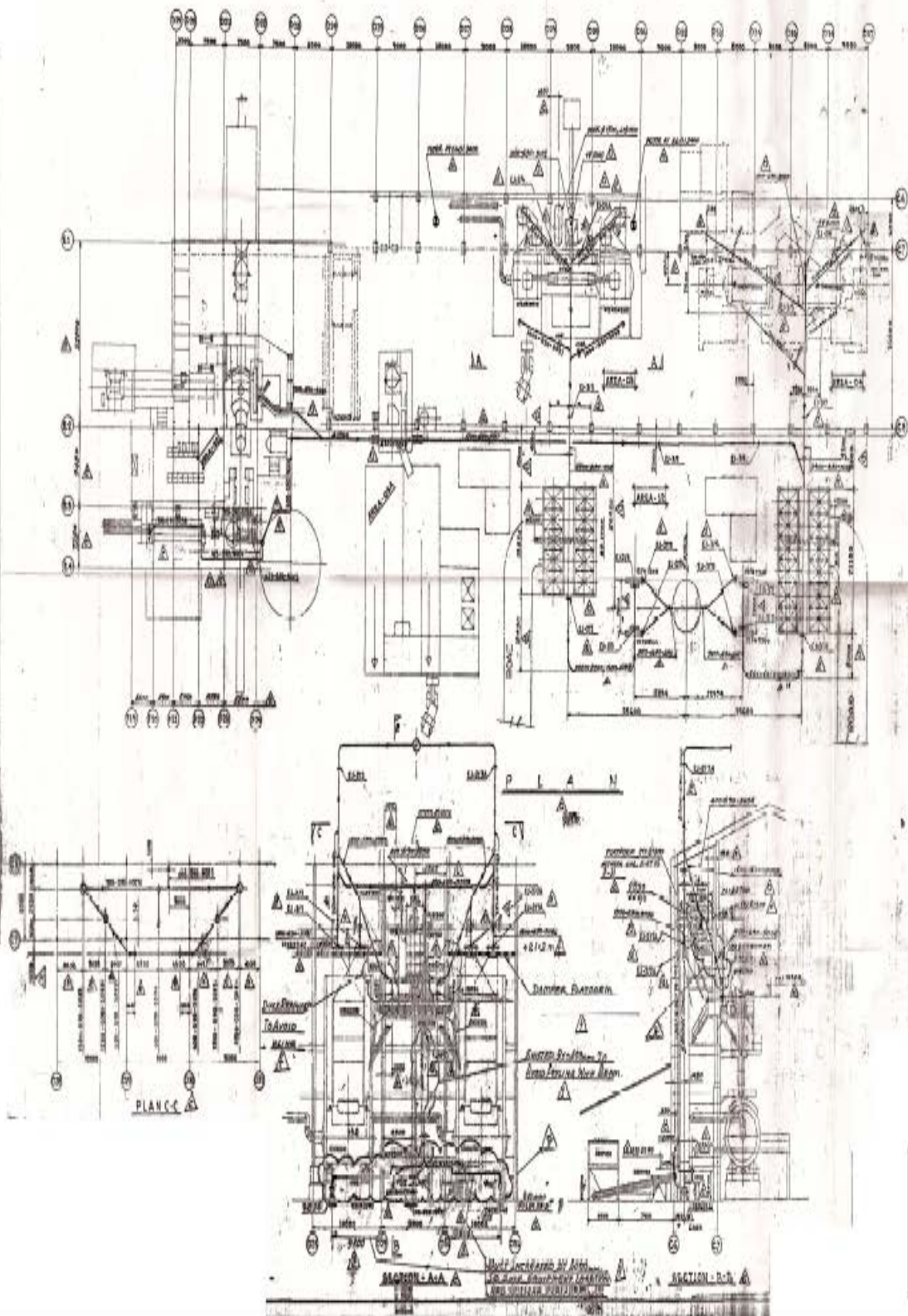
NOTE(S):

1. The Bidder/Contractor is responsible for removal/dismantling of the Dampers (installed for the ducts) from all heights/locations.
2. All tools/tackles & consumables required for the dismantling is in Contractor's scope.
3. Parts found damaged during overhauling /servicing shall be arranged/procured by the Owner.





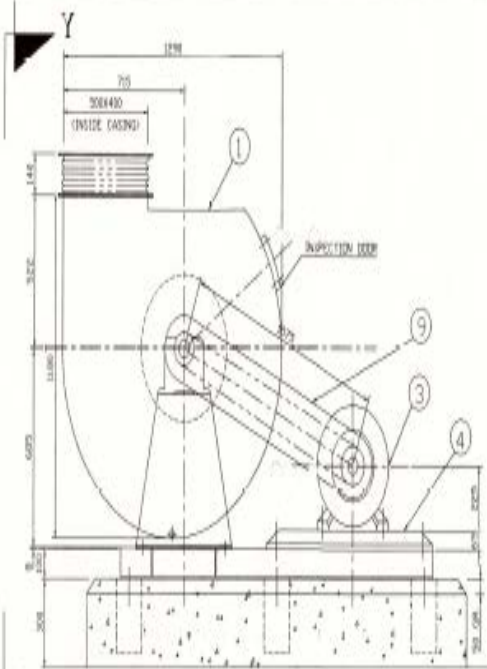




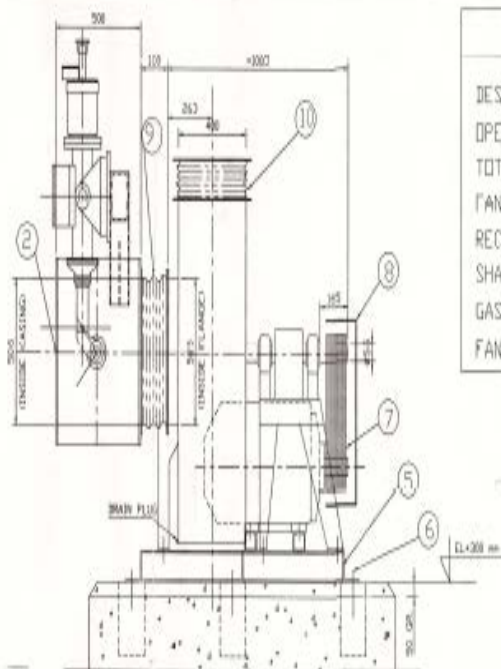
NOTE:
 1. ALL DIMENSIONS ARE IN METERS.

RAMSAY GROUP LIMITED
 1001 GILBERT STREET
 1001 GILBERT STREET
 1001 GILBERT STREET





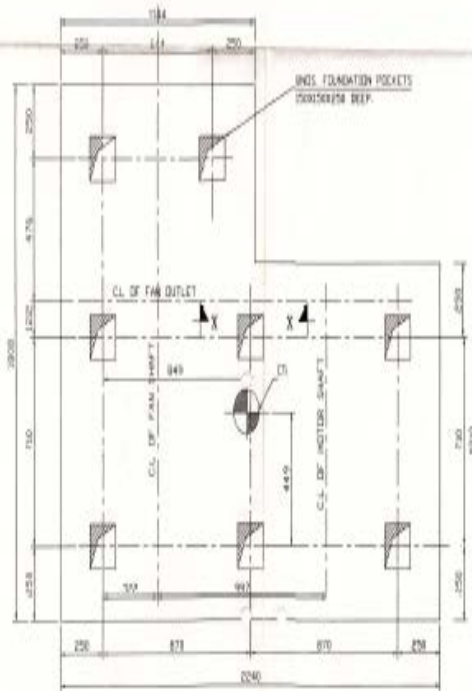
ELEVATION



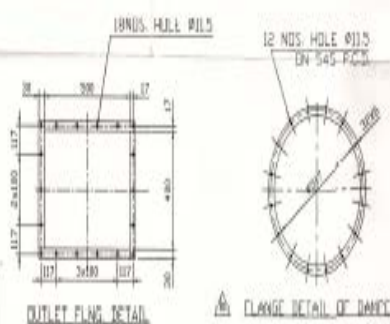
-Y-Y-

TECHNICAL SPEC:

DESIGN VOLUME	6.7m ³ /S.
OPERATING TEMP.	110°C
TOTAL PR.	300 mmWg
FAN SPEED	2050 R.P.M.
RECOMMENDED MOTOR RATING	37 KW/4POLE
SHAFT POWER AT OP.TEMP.	28.75 KW
GAS DENSITY AT 20°C	1.2 KG/m ³
FAN EFFICIENCY	70 %

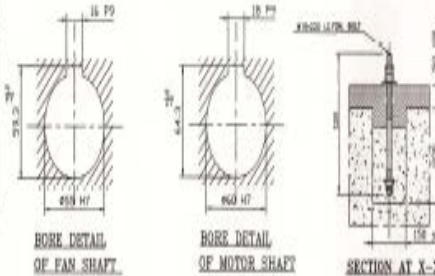


FOUNDATION DETAIL



OUTLET FLANGE DETAIL

DAMPER DETAIL OF DAMPER



SHAFT DETAIL OF FAN SHAFT

SHAFT DETAIL OF MOTOR SHAFT

SECTION AT X-X

NOTE:-

1. BEARINGS ARE GREASE LUBRICATED THROUGH GREASE NIPPLE FITTED ON PLUMMER BLOCK.
2. PAINTING: HEAT RESISTENT PAINT
3. DAMPER WILL BE SUPPORTED SEPARATELY.

1	OUTLET FLANGE	NON-METALIC
2	OUTLET PIPING	PPH
3	BELT GUARD	ALUMINUM
4	V-BELT (ANTI STATIC)	3000/10000
5	FOUNDATION BELT PULLEY	ML/15/2002-A
6	COMMON BASE FRAME	ML/15/2002-A
7	SLIP RAIL	ML/15/2002-A
8	MOTOR	37KW/4POLE/2050RPM/4
9	ROTTERLY DAMPER/SHUTTER	ML/15/2002-A
10	CENTRIFUGAL FAN	ML/15/2002-A

STATIC LOAD FOR FAN & ACCESSORIES- 718 KG.(APPROX.)

DYNAMIC LOAD FOR FAN- 80 KG.(APPROX.)
(DUE TO ROTOR.)

DESIGNED BY: [Signature]
CHECKED BY: [Signature]

DATE: 10/10/2010

Fig No. 10/10/2010

DO NOT SCALE THIS DRAWING

PART NO	DESCRIPTION	QTY	MATERIAL	REMARKS
1	Body	1	MS IS 2062 16 TH.	
2	Flap	1	MS IS 2062 16 TH.	
3	Stopper	2	MS IS 2062	
4	Seal	-	SS-316 0.7 TH.	
5	Flange	2	MS IS 2062 16 TH.	
6	Shaft-End Shaft	2	Carbon Steel ; EN-8/C-45	
7	Pipe	1	MS/EN	
8	Bearing	2	'SKF'/'FAG' Make	
9	Spring	2	Spring Steel	
10	Gland Housing Cover	2	MS	
11	Gland Packing	-	Non Asbestos	
12	Gland Housing	2	MS	
13	Drive Lever	1	MS	
14	Linear Actuator	1	'UNITED TECHNOMECH' Make	
15	Below	1	Rubber Coated Nylon Fabric	
16	Motor	1	'NCC'/'BBL' Make	
17	Interlocking Limit Switch	1	'JAI BALAJI' Make	
18	Ext. Travel Limit Switch	2	'JAI BALAJI' Make	
19	Calibration Scale	1	SS-304	

SUB. NO.	TAG. NO.	QTY.
1102 & 1108	PV 100	1 NO.
2102 & 2108	PV 2000	1 NO.

NOTE

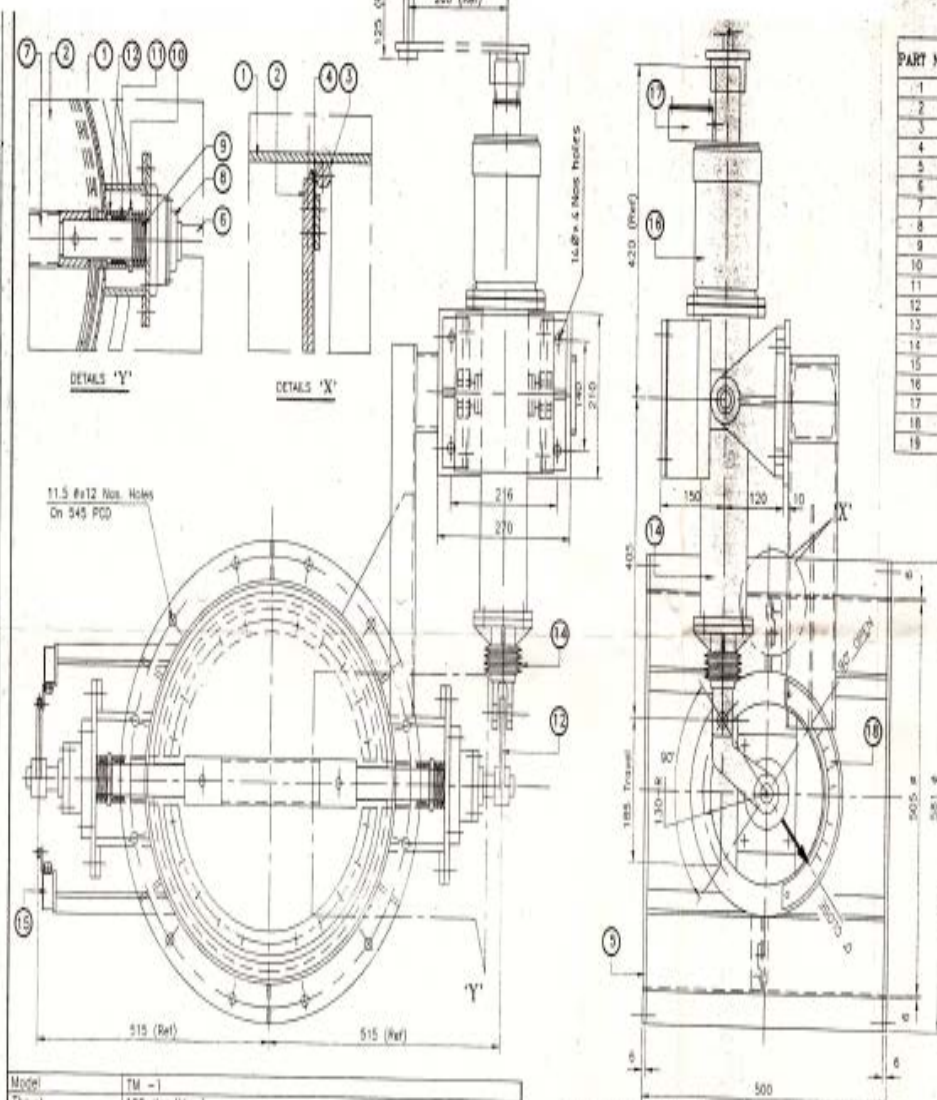
- APPLICATION :- FAN INLET DAMPER HANDLING CLEAN AIR
- OPERATING TEMP :- 110° C
- OPERATING PRESSURE :- (-) 500 MM WG
- SEALING EFFICIENCY :- 99.8 ACA
- PAINING :- ALL STEEL PARTS WILL BE PAINTED WITH 2 COATS OF HEAT RESISTANT PAINT (ALUMINIUM) SUITABLE FOR THE OPERATING TEMPERATURE.
- PRESSURE DROP :- 80 MM WG/MIN WHEN DAMPER IS FULLY OPEN.
- IN COMBINATION WITH BUTTERFLY DAMPER INSTALL THE SUITABLE QUALITY EXPANSION JOINT TO PROVIDE A POINT OF FLEXIBILITY IN FLUE DUCTS TO RELIEVE STRESS CAUSED BY SYSTEM MOVEMENT.

TOTAL WEIGHT : 185 Kgs (Approx)

HINDUSTAN COPPER LIMITED
UNIT : GUJARAT COPPER PROJECT

TITLE : GENERAL ASSEMBLY OF MOTORISED DAMPER
PV1000 & PV2000

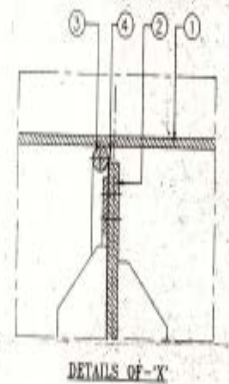
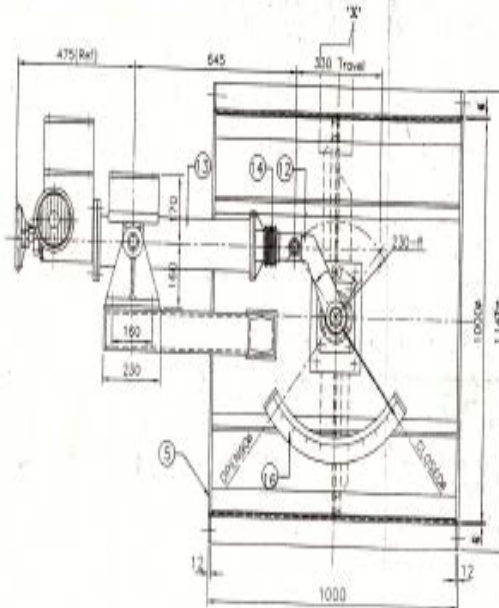
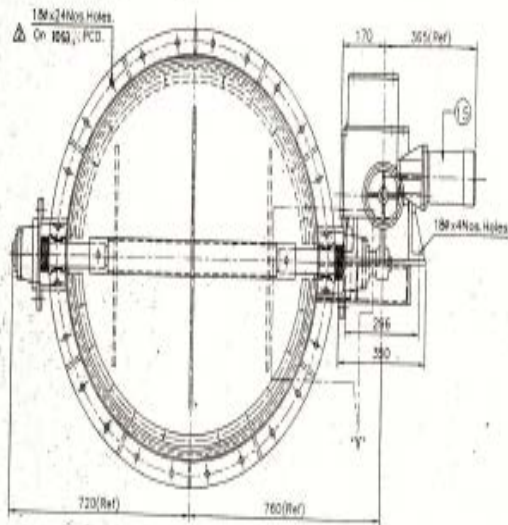
Des No. : HCU/GCP/12-BPM/001 Rev. 0



Model	TM -1
Thrust	100 Kgs (Max.)
Stroke	200 MM/Adjustable
Speed	75MM/Sec (Approx.)
Motor	0.5HP/0.37KW/4 Pole, 415V±10%, 50Hz±5% (Combined variation ±10%) 3 Phase 'AC' supply cage 'TEFC' Class 'F' insulation with temp. rise limited to class 'B' IP-55 Enclosure suitable upto 50° C Ambient Temp. (Make:- 'NCC'/'BBL') Along With canopy
Overload Protection	Provided with 2Nos. Limit Switches having 2NO+2NC Potential Free contacts rated for 230 V, 5 Amps.
Travel Adjustment	Provided with 2Nos. External Limit Switches having 2NO+2NC Potential Free contacts rated for 230 V, 5 Amps.
Manual Operation	Provided With Interlocking facility with 1 No. Limit Switch 1NO+1NC Contacts rated for 230 V, 5 Amps.

TECHNICAL SPECIFICATIONS.

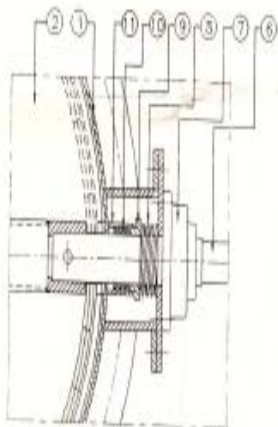
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NOTE

- 1) APPLICATION :- INSTALLED IN DUST LOADED FLUE GAS PATH FROM ANODE FURNACE REGULATING DUTY TO MAINTAIN DESIRED PRESSURE AT TAP OFF POINT ON ANODE FURNACE.
- 2) OPERATING TEMP :- 450°C
- 3) OPERATING PRESSURE :- (-) 200 MM HG
- 4) SEALING EFFICIENCY :- 99%
- 5) PAINTING :- ALL STEEL PARTS WILL BE PAINTED WITH 2 COATES OF HEAT RES. JANT PAINT (ALUMINUM) SUITABLE FOR THE OPERATING TEMPERATURE.
- 6) PRESSURE DROP 25mm Wg (max) WHEN THE DAMPER IS FULLY OPEN
- 7) IN COMBINATION WITH BUTTERFLY DAMPER INSTALL THE SUITABLE QUALITY EXPANSION JOINT TO PROVIDE A POINT OF FLEXIBILITY IN FLUE DUCTS TO RELIEVE STRESS CAUSED BY SYSTEM MOVEMENT.
- 8) WIRING DIAGRAM REFER OUR ENG. NO. - UTE 01000 71.

SUB NO. : 4016.
TAG NO. : PVO250, PVO550.
P.O. NO. : PVO.317.0018 00. DATE : 10/09/07.
QUANTITY : 2 NOS.
TOTAL WEIGHT : 360 KGS. (APPROX)



DETAILS OF 'Y'

PART NO	DESCRIPTION	QTY	MATERIAL	REMARKS
16	Calibration Scale	1	SS-304	
15	Motor	1	KEC/MBL Make	
14	Bellev	1	Rubber Coated Nylon fabric	
13	Linear Actuator (LWR-2)	1	UNITED TECH/NOVECH Make	
12	Drive Lever	1	MS	
11	Gland Housing	2	MS	
10	Gland Packing	-	Non Asbestos	
9	Gland Housing Cover	2	MS	
8	Spring	2	Spring Steel	
7	Spherical bearing (GE 60 ES)	2	Self / FAG Make	
6	Stub-End Shaft (60 Ø)	2	Carbon Steel EN-8/C-45	
5	Flange	2	RQ15-2002 Grade 1/12 Thk	
4	Seal	-	SS-316 0.7 Thk	
3	Stopper	2	MS	
2	Flag	1	RQ15-2002 Grade 1/12 Thk	
1	Body	1	RQ15-2002 Grade 1/6 Thk	

Model	TWR-2
Thrust	750 Kgs (Max)
Stroke	350 MM (Adjustable)
Speed	4 MM/Sec (Approx)
Motor	0.5HP/0.37KW 4 Pole, 415V±10%, 50Hz 50% (Combined with 100% 3 Phase AC Supply squirrel cage "IEPC" Class F insulation with temp rise limited to 80°C B P=55 (Enclosure available upto 50° C Ambient temp (Make: OBL / KEC) Along with Consigny.
Overload Protection	Provided with 2 Nos. Limit Switches having 2NO+2NC Potential free contacts rated for 230V, 5amps.
Travel Adjustment	As
Position Transmitter	Provided with 135 Ohms, 3Watts rating (Resistor type)
Signal Operation	Provided with Handwheel in case of power failure.

TECHNICAL SPECIFICATIONS

HINDUSTAN COPPER LIMITED
UNIT: GUJARAT COPPER PROJECT

TITLE: GA OF BUTTERFLY DAMPER
FOR PVO250 & PVO550
Dwg. No. - HCL/GCP/13-BHMD03

	CABLE TAG NO.	FROM	THRU/ TR.NO.	TO	CABLE DESCRIPTION				
					NO. OF CORES	SIZE (MM ²)	TYPE	VOLTS	LENGTH (M)
01	SYSTEM - III 3FC-RFM2008P	MCC-3 (1A3T4S3)	TRAY SIII-TR-02/ 04/11	ROTARY FEEDER - 7 RFM2007M	3/C	4	AYWY	1100	40
02	3FD-RFM2006C	MCC-3 (1A3T4S3)	TRAY SIII-TR-01/ 04/11	LOCAL PUSH BUTTON STATION	5/C	2.5	YWY	1100	40
03	3FE-RFM2008P	MCC-3 (1A3T4S3)	TRAY SIII-TR-02/ 03/10	ROTARY FEEDER - 8 RFM2008M	3/C	4	AYWY	1100	30
04	3FE-RFM2006C	MCC-3 (1A3T4S3)	TRAY SIII-TR-01/ 03/11	LOCAL PUSH BUTTON STATION	5/C	2.5	YWY	1100	30
05	2FC-RFM2005P	MCC-3 (1A3T4S3)	TRAY SIII-TR-02/ 03/09	ROTARY FEEDER - 5 RFM2005M	3/C	4	AYWY	1100	30
06	2FC-RFM2005C	MCC-3 (1A3T4S3)	TRAY SIII-TR-01/ 03/08	LOCAL PUSH BUTTON STATION	5/C	2.5	YWY	1100	30
07	2FB-RFM2004P	MCC-3 (1A3T4S3)	TRAY SIII-TR-02/ 03/08	ROTARY FEEDER - 4 RFM2004M	3/C	4	AYWY	1100	25
08	2FB-RFM2004C	MCC-3 (1A3T4S3)	TRAY SIII-TR-01/ 03/08	LOCAL PUSH BUTTON STATION	5/C	2.5	YWY	1100	25
09	1FD-RFM2003P	MCC-3 (1A3T4S3)	TRAY SIII-TR-02/ 03/07	ROTARY FEEDER - 3 RFM2003M	3/C	4	AYWY	1100	25
10	1FD-RFM2003C	MCC-3 (1A3T4S3)	TRAY SIII-TR-01/ 03/07	LOCAL PUSH BUTTON STATION	5/C	2.5	YWY	1100	25
11	1FC-RFM2002P	MCC-3 (1A3T4S3)	TRAY SIII-TR-02/ 03/06	ROTARY FEEDER - 2 RFM2002M	3/C	4	AYWY	1100	30
12	1FC-RFM2002C	MCC-3 (1A3T4S3)	TRAY SIII-TR-01/ 03/06	LOCAL PUSH BUTTON STATION	5/C	2.5	YWY	1100	30
13	1FB-RFM2001P	MCC-3 (1A3T4S3)	TRAY SIII-TR-02/ 03/05	ROTARY FEEDER - 1 RFM2001M	3/C	4	AYWY	1100	30

SL. NO.	CABLE TAG NO.	FROM	THRU/ TR. NO.	TO	CABLE DESCRIPTION				
					NO. OF CORES	SIZE (MM ²)	TYPE	VOLTS	LENGTH (M)
14	1FB-RFM2001C	MCC-3 (1A3T4S3)	TRAY SIII-TR-01/ 03/05	LOCAL PUSH BUTTON STATION	5/C	2.5	YWW	1100	30
15	4RC-12FNM2300P	MCC-3 (1A3T4S3)	H.T. TRAY H.T.TRENCH	ID FAN INLET DAMPER - 2 12FNM2300M	3/C	4	AYWY	1100	25
16	4RC-12FNM2300C	MCC-3 (1A3T4S3)	H.T. TRAY H.T.TRENCH 04/11	LOCAL PUSH BUTTON STATION	5/C	2.5	YWW	1100	25
17	4FE-RFM2010P	MCC-3 (1A3T4S3)	TRAY SIII-TR-02/ 04/14	ROTARY FEEDER - 10 RFM2010M	3/C	4	AYWY	1100	35
18	4FE-RFM2010C	MCC-3 (1A3T4S3)	TRAY SIII-TR-01/ 04/14	LOCAL PUSH BUTTON STATION	5/C	2.5	YWW	1100	35
19	4FD-RFM2009P	MCC-3 (1A3T4S3)	TRAY SIII-TR-02/ 04/13	ROTARY FEEDER - 9 RFM2009M	3/C	4	AYWY	1100	35
20	4FD-RFM2009C	MCC-3 (1A3T4S3)	TRAY SIII-TR-01/ 04/13	LOCAL PUSH BUTTON STATION	5/C	2.5	YWW	1100	35
21	4FC-RFM2008P	MCC-3 (1A3T4S3)	TRAY SIII-TR-02/ 04/12	ROTARY FEEDER - 8 RFM2008M	3/C	4	AYWY	1100	40
22	4FC-RFM2008C	MCC-3 (1A3T4S3)	TRAY SIII-TR-01/ 04/12	LOCAL PUSH BUTTON STATION	5/C	2.5	YWW	1100	40
23	4FB-RCFM2000P	MCC-3 (1A3T4S3)	TRAY SIII-TR-02/ 02A	RE-CIRCULATION FAN RCFM2000M	3/C	70	AYWY	1100	60
24	4FB-RCFM2000C	MCC-3 (1A3T4S3)	TRAY SIII-TR-01/ 02A	LOCAL PUSH BUTTON STATION	5/C	2.5	YWW	1100	60
25	3FE-SCM2007P	MCC-3 (1A3T4S3)	TRAY SIII-TR-02/ 04	SCREW CONVEYOR - 7 SCM2007M	3/C	10	AYWY	1100	30
26	3FE-SCM2007C	MCC-3 (1A3T4S3)	TRAY SIII-TR-01/ 04	LOCAL PUSH BUTTON STATION	5/C	2.5	YWW	1100	30
27	3FD-RFM2007P	MCC-3	TRAY SIII-TR-02/ 03	SCREW CONVEYOR - 1 SCM2001M	3/C	10	AYWY	1100	25

S. NO.	CABLE TAG NO.	FROM	THRU/ TR.NO.	TO	CABLE DESCRIPTION				
					NO. OF CORES	SIZE (MM ²)	TYPE	VOLTS	LENGTH (M)
28	3FD-RFM2007C	MCC-3	TRAY SIII-TR-01/ 03	LOCAL PUSH BUTTON STATION	5/C	2.5	YWW	1100	25
29	3RG-12FNM2200P	MCC-3 (1A3T4S3)	H.T TRAY/ H.T TRENCH	ID FAN INLET DAMPER - 1 12FNM2200M	3/C	4	AYWW	1100	40
30	3RG-12FNM2200C	MCC-3 (1A3T4S3)	H.T TRAY/ H.T TRENCH	LOCAL PUSH BUTTON STATION	5/C	2.5	YWW	1100	40
31	3RE-PVM2000P	MCC-3 (1A3T4S3)	TRAY SII-TR-02/ 02/A	RE-CIRCULATION FAN INLET DAMPER PVM2000M	3/C	4	AYWW	1100	55
32	3RE-PVM2000C	MCC-3 (1A3T4S3)	TRAY SII-TR-01/ 02A	LOCAL PUSH BUTTON STATION	5/C	2.5	YWW	1100	55
33	3RC-SCM2006P	MCC-3 (1A3T4S3)	TRAY SIII-TR-02/ 03	SCREW CONVEYOR - 8 SCM2006M	3/C	4	AYWW	1100	25
34	3RC-SCM2006C	MCC-3 (1A3T4S3)	TRAY SIII-TR-01/ 03	LOCAL PUSH BUTTON STATION	5/C	2.5	YWW	1100	25
35	4RF-PVM0550P	MCC-3 (1A3T4S3)	TRAY SIII-TR-03/ 23/24	RE-CIRCULATION FAN INLET DAMPER PVM0550M	3/C	4	AYWW	1100	35
36	4RF-PVM0550C	MCC-3 (1A3T4S3)	TRAY SIII-TR-03/ 24	LOCAL PUSH BUTTON STATION	5/C	2.5	YWW	1100	35
37	1RC-RFM2012P	MCC-3 (1A3T4S3)	TRAY SIII-TR-02/ 04/16	ROTARY FEEDER - 12 RFM2012M	3/C	4	AYWW	1100	40
38	1RC-RFM2012C	MCC-3 (1A3T4S3)	TRAY SIII-TR-01/ 04/16	LOCAL PUSH BUTTON STATION	5/C	2.5	YWW	1100	40
39	1RB-RFM2011P	MCC-3 (1A3T4S3)	TRAY SIII-TR-02/ 04/15	ROTARY FEEDER - 11 RFM2011M	3/C	4	AYWW	1100	35
40	1RB-RFM2011C	MCC-3 (1A3T4S3)	TRAY SIII-TR-01/ 04/15	LOCAL PUSH BUTTON STATION	5/C	2.5	YWW	1100	35
41	2RG-SCM2005P	MCC-3 (1A3T4S3)	TRAY SIII-TR-02/ 03	SCREW CONVEYOR - 5 SCM2005M	3/C	4	AYWW	1100	30

SL. NO.	CABLE TAG NO.	FROM	THRU/ TR. NO.	TO	CABLE DESCRIPTION				
					NO. OF CORES	SIZE (MM ²)	TYPE	VOLTS	LENGTH (M)
42	2RG-SCM2005C	MCC-3 (1A3T4S3)	TRAY SIII-TR-01/03	LOCAL PUSH BUTTON STATION	5/C	2.5	YWW	1100	30
43	2RF-SCM2004P	MCC-3 (1A3T4S3)	TRAY SIII-TR-02/03	SCREW CONVEYOR - 4 SCM2004M	3/C	4	AYWY	1100	25
44	2RF-SCM2004C	MCC-3 (1A3T4S3)	TRAY SIII-TR-01/03	LOCAL PUSH BUTTON STATION	5/C	2.5	YWW	1100	25
45	2RE-SCM2003P	MCC-3 (1A3T4S3)	TRAY SIII-TR-02/03	SCREW CONVEYOR - 3 SCM2003M	3/C	4	AYWY	1100	20
46	2RE-SCM2003C	MCC-3 (1A3T4S3)	TRAY SIII-TR-01/03	LOCAL PUSH BUTTON STATION	5/C	2.5	YWW	1100	20
47	2RD-SCM2002P	MCC-3 (1A3T4S3)	TRAY SIII-TR-02/03	SCREW CONVEYOR - 2 SCM2002M	3/C	4	AYWY	1100	20
48	2RD-SCM2002C	MCC-3 (1A3T4S3)	TRAY SIII-TR-01/03	LOCAL PUSH BUTTON STATION	5/C	2.5	YWW	1100	20
49	3FB-JBMH-01P	MCC-3 (1A3T4S3)	TRAY SIII-TR-02/02A	MAIN HEATER JUNCTION BOX	4/C	120	AYWY	1100	25
50	03JBMH-01P	MAIN HEATER JUNCTION BOX	TRAY SIII-TR-02A/03/17A	HEATER JUNCTION BOX 3JBH1	3/C	6	AYWY	1100	15
51	03JBMH-01C	MAIN HEATER JUNCTION BOX	TRAY SIII-TR-02A/03/17A	THERMOSTAT	3/C	2.5	YWW	1100	10
52	3JBH1-P1 TO P4	HEATER JUNCTION BOX	TRAY SIII-TR-17A	4 X 1 KW HOPPER HEATER	4 X 3/C	4	AYWY	1100	4 X 7 = 30
53	03JBMH-02P	MAIN HEATER JUNCTION BOX	TRAY SIII-TR-02A/03/18A	HEATER JUNCTION BOX 3JBH2	3/C	6	AYWY	1100	10
54	03JBMH-02C	MAIN HEATER JUNCTION BOX	TRAY SIII-TR-02A/03/18A	THERMOSTAT	3/C	2.5	YWW	1100	10

SL. NO.	CABLE TAG NO.	FROM	THRU/ TR.NO.	TO	CABLE DESCRIPTION				
					NO. OF CORES	SIZE (MM ²)	TYPE	VOLTS	LENGTH (M)
56	3JBH2-P1 TO P4	HEATER JUNCTION BOX	TRAY SIII-TR-18A	4 X 1 KW HOPPER HEATER	4 X 3/C	4	AYWY	1100	4 X 7 = 30
56	03JBMH-03P	MAIN HEATER JUNCTION BOX	TRAY SIII-TR-02A/ 03/19A	HEATER JUNCTION BOX 3JBH3	3/C	6	AYWY	1100	5
57	03JBMH-03C	MAIN HEATER JUNCTION BOX	TRAY SIII-TR-02A/ 03/19A	THERMOSTAT	3/C	2.5	YWY	1100	10
58	3JBH3-P1 TO P4	HEATER JUNCTION BOX	TRAY SIII-TR-19A	4 X 1 KW HOPPER HEATER	4 X 3/C	4	AYWY	1100	4 X 7 = 30
59	03JBMH-04B	MAIN HEATER JUNCTION BOX	TRAY SIII-TR-02A/ 03/20A	HEATER JUNCTION BOX 3JBH4	3/C	6	AYWY	1100	5
60	03JBMH-04C	MAIN HEATER JUNCTION BOX	TRAY SIII-TR-02A/ 03/20A	THERMOSTAT	3/C	2.5	YWY	1100	10
61	3JBH4-P1 TO P4	HEATER JUNCTION BOX	TRAY SIII-TR-20A	4 X 1 KW HOPPER HEATER	4 X 3/C	4	AYWY	1100	4 X 7 = 30
62	03JBMH-05P	MAIN HEATER JUNCTION BOX	TRAY SIII-TR-02A/ 03/21A	HEATER JUNCTION BOX 3JBH5	3/C	6	AYWY	1100	10
63	03JBMH-05C	MAIN HEATER JUNCTION BOX	TRAY SIII-TR-02A/ 03/21A	THERMOSTAT	3/C	2.5	YWY	1100	10
64	3JBH5-P1 TO P4	HEATER JUNCTION BOX	TRAY SIII-TR-21A	4 X 1 KW HOPPER HEATER	4 X 3/C	4	AYWY	1100	4 X 7 = 30
65	03JBMH-06P	MAIN HEATER JUNCTION BOX	TRAY SIII-TR-02A/ 03/22A	HEATER JUNCTION BOX 3JBH6	3/C	6	AYWY	1100	15
66	03JBMH-06C	MAIN HEATER JUNCTION BOX	TRAY SIII-TR-02A/ 03/22A	THERMOSTAT	3/C	2.5	YWY	1100	10

SL. NO.	CABLE TAG NO.	FROM	THRU/ TR.NO.	TO	CABLE DESCRIPTION				
					NO. OF CORES	SIZE (MM ²)	TYPE	VOLTS	LENGTH (M)
67	3JBH6-P1 TO P4	HEATER JUNCTION BOX	TRAY SIII-TR-22A	4 X 1 KW HOPPER HEATER	4 X 3/C	4	AYWY	1100	4 X 7 = 30
68	03JBMH-07P	MAIN HEATER JUNCTION BOX	TRAY SIII-TR-02A 04/17	HEATER JUNCTION BOX 3JBH7	3/C	6	AYWY	1100	25
69	03JBMH-07C	MAIN HEATER JUNCTION BOX	TRAY SIII-TR-02A/ 04/17	THERMOSTAT	3/C	2.5	YWY	1100	10
70	3JBH7-P1 TO P4	HEATER JUNCTION BOX	TRAY SIII-TR-17	4 X 1 KW HOPPER HEATER	4 X 3/C	4	AYWY	1100	4 X 7 = 30
71	03JBMH-08P	MAIN HEATER JUNCTION BOX	TRAY SIII-TR-02A/ 04/18	HEATER JUNCTION BOX 3JBH8	3/C	6	AYWY	1100	20
72	03JBMH-08C	MAIN HEATER JUNCTION BOX	TRAY SIII-TR-02A/ 04/18	THERMOSTAT	3/C	2.5	YWY	1100	10
73	3JBH8-P1 TO P4	HEATER JUNCTION BOX	TRAY SIII-TR-18	4 X 1 KW HOPPER HEATER	4 X 3/C	4	AYWY	1100	4 X 7 = 30
74	03JBMH-09P	MAIN HEATER JUNCTION BOX	TRAY SIII-TR-02A/ 04/19	HEATER JUNCTION BOX 3JBH9	3/C	6	AYWY	1100	20
75	03JBMH-09C	MAIN HEATER JUNCTION BOX	TRAY SIII-TR-02A/ 04/19	THERMOSTAT	3/C	2.5	YWY	1100	10
76	3JBH9-P1 TO P4	HEATER JUNCTION BOX	TRAY SIII-TR-19	4 X 1 KW HOPPER HEATER	4 X 3/C	4	AYWY	1100	4 X 7 = 30
77	03JBMH-10P	MAIN HEATER JUNCTION BOX	TRAY SIII-TR-02A/ 04/20	HEATER JUNCTION BOX 3JBH10	3/C	6	AYWY	1100	20
78	03JBMH-10C	MAIN HEATER JUNCTION BOX	TRAY SIII-TR-02A/ 04/20	THERMOSTAT	3/C	2.5	YWY	1100	10

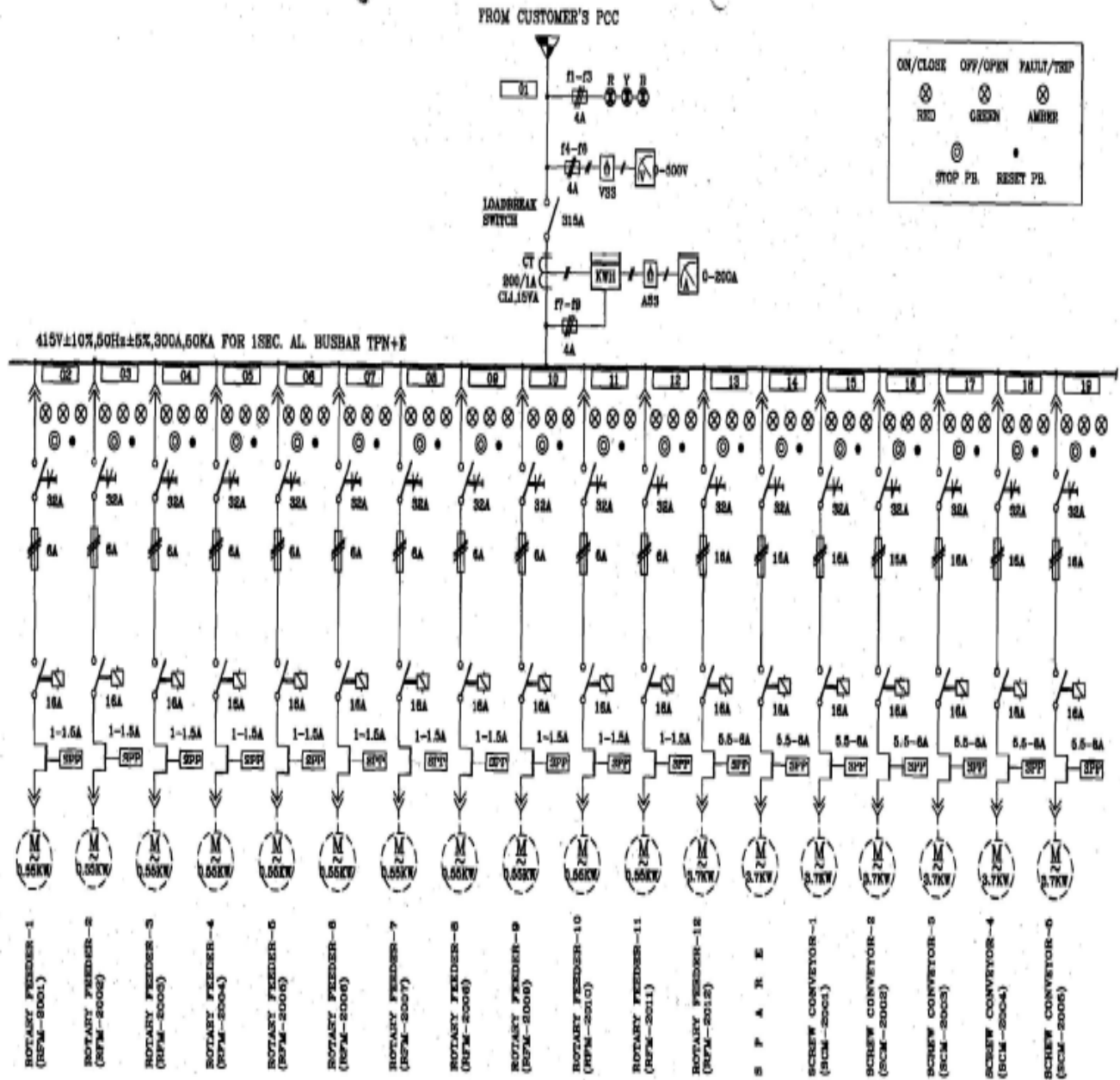
SL. NO.	CABLE TAG NO.	FROM	THRU/ TR.NO.	TO	CABLE DESCRIPTION				
					NO. OF CORES	SIZE (MM ²)	TYPE	VOLTS	LENGTH (M)
79	3JBH10-P1 TO P4	HEATER JUNCTION BOX	TRAY SIII-TR-20	4 X 1 KW HOPPER HEATER	4 X 3/C	4	AYWY	1100	4 X 7 = 30
80	03JBMH-11P	MAIN HEATER JUNCTION BOX	TRAY SIII-TR-02A/ 04/20	HEATER JUNCTION BOX 3JBH11	3/C	6	AYWY	1100	25
81	03JBMH-11C	MAIN HEATER JUNCTION BOX	TRAY SIII-TR-02A/ 04/21	THERMOSTAT	3/C	2.5	YWY	1100	10
82	3JBH11-P1 TO P4	HEATER JUNCTION BOX	TRAY SIII-TR-21	4 X 1 KW HOPPER HEATER	4 X 3/C	4	AYWY	1100	4 X 7 = 30
83	03JBMH-12P	MAIN HEATER JUNCTION BOX	TRAY SIII-TR-02A/ 04/22	HEATER JUNCTION BOX 3JBH12	3/C	6	AYWY	1100	30
84	03JBMH-12C	MAIN HEATER JUNCTION BOX	TRAY SIII-TR-02A/ 04/22	THERMOSTAT	3/C	2.5	YWY	1100	10
85	3JBH12-P1 TO P4	HEATER JUNCTION BOX	TRAY SIII-TR-22	4 X 1 KW HOPPER HEATER	4 X 3/C	4	AYWY	1100	4 X 7 = 30
86	1F-PLC-3-C1	MCC -3 (IA3T4S3)	TRAY	PLC - 3	19/C	1.5	YWY	1100	20
87	1F-PLC-3-C2	MCC -3 (IA3T4S3)	TRAY	PLC - 3	16/C	1.5	YWY	1100	15
88	2F-PLC-3-C1	MCC -3 (IA3T4S3)	TRAY	PLC - 3	10/C	1.5	YWY	1100	20
89	2F-PLC-3-C2	MCC -3 (IA3T4S3)	TRAY	PLC - 3	10/C	1.5	YWY	1100	15
90	3F-PLC-3-C1	MCC -3 (IA3T4S3)	TRAY	PLC - 3	19/C	1.5	YWY	1100	20
91	3F-PLC-3-C2	MCC -3 (IA3T4S3)	TRAY	PLC - 3	16/C	1.5	YWY	1100	15
92	4F-PLC-3-C1	MCC -3 (IA3T4S3)	TRAY	PLC - 3	19/C	1.5	YWY	1100	20
93	4F-PLC-3-C2	MCC -3 (IA3T4S3)	TRAY	PLC - 3	16/C	1.5	YWY	1100	15

S. NO.	CABLE TAG NO.	FROM	THRU/ TR. NO.	TO	CABLE DESCRIPTION				
					NO. OF CORES	SIZE (MM ²)	TYPE	VOLTS	LENGTH (M)
94	4R-PLC-3-C1	MCC -3 (IA3T4S3)	TRAY	PLC - 3	18/C	1.5	YWY	1100	20
95	4R-PLC-3-C2	MCC -3 (IA3T4S3)	TRAY	PLC - 3	16/C	1.5	YWY	1100	15
96	3R-PLC-3-C1	MCC -3 (IA3T4S3)	TRAY	PLC - 3	19/C	1.5	YWY	1100	20
97	3R-PLC-3-C2	MCC -3 (IA3T4S3)	TRAY	PLC - 3	16/C	1.5	YWY	1100	15
98	2R-PLC-3-C1	MCC -3 (IA3T4S3)	TRAY	PLC - 3	19/C	1.5	YWY	1100	20
99	2R-PLC-3-C2	MCC -3 (IA3T4S3)	TRAY	PLC - 3	16/C	1.5	YWY	1100	15
100	1R-PLC-3-C1	MCC -3 (IA3T4S3)	TRAY	PLC - 3	10/C	1.5	YWY	1100	20
101	1R-PLC-3-C2	MCC -3 (IA3T4S3)	TRAY	PLC - 3	10/C	1.5	YWY	1100	15
102	4FF-LDB2	MCC -3 (IA3T4S3)	TRAY	LDB-2	4C	16	YWY	1100	30
103	2FF-JBS-P1	MCC -3 (IA3T4S3)	TRAY	03-JBS-002	3C	2.5	YWY	1100	35

SL. NO.	EQUIPMENT NO.	SERVICES	QTY.	RATED		OPERATING						REMARKS
				NORMAL	STANDBY	MECH.	EFFC.	KW	P.F	KVA	KVAR	
				(KW)	(KW)	(KW)	(%)					
SYSTEM - III												
1 a)	RFM-2001	ROTARY FEEDER BELOW HOPPER OF BAG HOUSE (BG 2000)	1	0.55	-	0.3	57	0.526	0.82	0.641	0.367	DOL FEEDER
b)	RFM-2002	- DO -	1	0.55	-	0.3	57	0.526	0.82	0.641	0.367	DOL FEEDER
c)	RFM-2003	- DO -	1	0.55	-	0.3	57	0.526	0.82	0.641	0.367	DOL FEEDER
d)	RFM-2004	- DO -	1	0.55	-	0.3	57	0.526	0.82	0.641	0.367	DOL FEEDER
e)	RFM-2005	- DO -	1	0.55	-	0.3	57	0.526	0.82	0.641	0.367	DOL FEEDER
f)	RFM-2006	- DO -	1	0.55	-	0.3	57	0.526	0.82	0.641	0.367	DOL FEEDER
g)	RFM-2007	- DO -	1	0.55	-	0.3	57	0.526	0.82	0.641	0.367	DOL FEEDER
h)	RFM-2008	- DO -	1	0.55	-	0.3	57	0.526	0.82	0.641	0.367	DOL FEEDER
i)	RFM-2009	- DO -	1	0.55	-	0.3	57	0.526	0.82	0.641	0.367	DOL FEEDER
j)	RFM-2010	- DO -	1	0.55	-	0.3	57	0.526	0.82	0.641	0.367	DOL FEEDER
k)	RFM-2011	- DO -	1	0.55	-	0.3	57	0.526	0.82	0.641	0.367	DOL FEEDER
l)	RFM-2012	- DO -	1	0.55	-	0.3	57	0.526	0.82	0.641	0.367	DOL FEEDER
2 a)	SCM-2001	SCREW CONVEYOR UNDER ROTARY FEEDER	1	3.7	-	2.8	0.72	2.8	0.85	3.294	1.736	DOL FEEDER
b)	SCM-2002	- DO -	1	3.7	-	2.8	0.72	2.8	0.85	3.294	1.736	DOL FEEDER
c)	SCM-2003	- DO -	1	3.7	-	2.8	0.72	2.8	0.85	3.294	1.736	DOL FEEDER

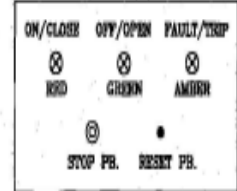
SL. NO.	EQUIPMENT NO.	SERVICES	QTY.	RATED		OPERATING						REMARKS
				NORMAL	STANDBY	MECH.	EFFC.	KW	P.F	KVA	KVAR	
				(KW)	(KW)	(KW)	(%)					
d)	SCM-2004	SCREW CONVEYOR UNDER ROTARY FEEDER	1	3.7	-	2.6	0.72	2.6	0.85	3.294	1.736	DOL FEEDER
e)	SCM-2006	- DO -	1	3.7	-	2.6	0.72	2.6	0.85	3.294	1.736	DOL FEEDER
f)	SCM-2007	- DO -	1	7.5	-	5	69	7.25	0.85	3.294	4.495	DOL FEEDER
3	RCFM-2000	RE-CIRCULATION FAN MOTOR	1	37	-	28.75	82.5	31.08	0.86	36.14	18.44	DOL FEEDER
4	PVM-2000	INLET DAMPER MOTOR OF RE-CIRCULATION PUMP	1	0.37	-	0.22	72	0.305	0.85	0.358	0.187	F/R FEEDER
5 a)	XVM-2000	INLET DAMPER MOTOR OF ID FAN	1	0.55	0.55	0.38	76	0.5	0.82	0.61	0.35	F/R FEEDER
b)	XVM-2300	- DO -	1	0.55	0.55	0.38	76	0.5	0.82	0.61	0.35	F/R FEEDER
6 a)	PVM-0250	REGULATING DAMPER	1	0.37	-	0.25	75	0.33	0.83	0.4	0.223	F/R FEEDER
b)	PVM-0550	- DO -	1	0.37	-	0.25	75	0.33	0.83	0.4	0.223	F/R FEEDER
7	3HH-01 TO 3HH-48	HOPPER HEATER	48	48 X 1	-	48	-	48	1	48	-	415 V FEEDER
8	-	LIGHTING LOAD TOTAL	-	8.5	-	-	-	8.5	0.9	9.44	4.11	415 V FEEDER
				128.31				117.107		128.649	41.462	

* FOR EFFICIENCY AND POWER FACTOR REFER TECHNICAL DATA SHEET OF RESPECTIVE MOTOR.

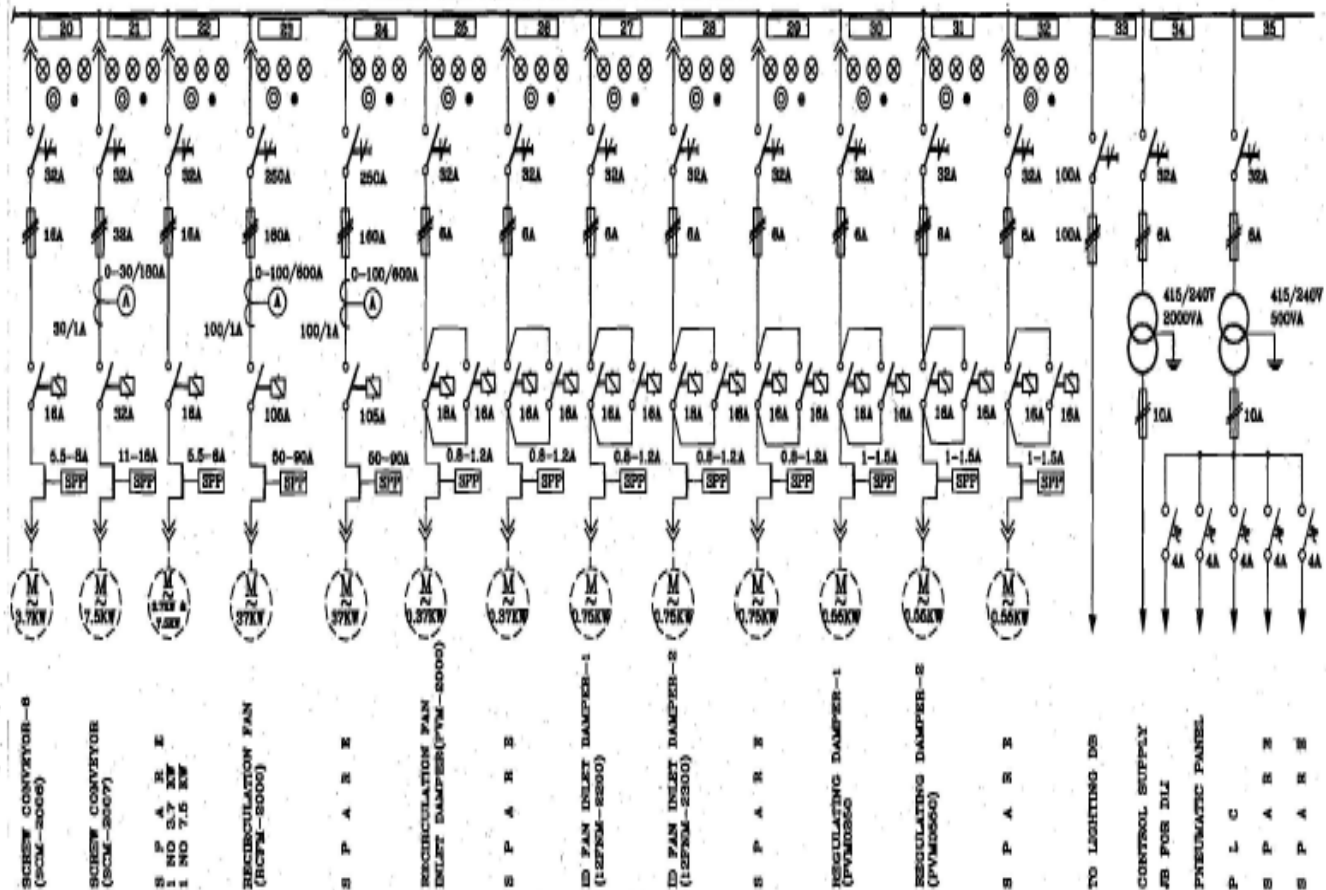


NOTES:

- 1) ALL SWITCHGEAR COMPONENTS SHALL COMPLY TO IEC DATA SHEET NO.8717-00-12-DS-0080-REV A & TYPE-C CO-ORDINATED
- 2) THE HORIZONTAL OVERLOAD RELAY SHALL BE WITH SINGLE PHASE PREVENTER
- 3) ALL MOTORS ABOVE 22KW SHALL BE PROVIDED WITH 22KW & 22KW



415V±10%, 50Hz±5%, 300A, 50KA FOR 1SEC. AL. BUSBAR TPN+E



PROJECT : GUJARAT COPPER PROJECT

DATA SHEET FOR INSTRUMENTATION FOR DEDUSTING SYSTEM - III

Doc. No. HCL/GCP/12-III/INST-02 Rev 0

No. of Sheets : 14

RESISTANCE TEMPERATURE DETECTOR WITH THERMOWELL / ADAPTER

SL. NO.	DESCRIPTION	
01	MAKE	NAGMAN
02	MATERIAL	PT - 100 WOUND ON CERAMIC
03	RANGE	0 TO 200 DEG.
04	TYPE	3 - WIRE MINERAL, INSULATED / DUPLEX
05	OUTER SHEATH AND DIA	21.8 MM, SS - 304
06	HEAD	DIE CASE AL. EPOXY PAINTED, WEATHER PROOF WITH SS CHAIN & BAKELITE BLOCK & NI PLATED BRASS TERMINALS
07	FILLING	94% MgO POWDER
08	PROBES: SHEATH OD	SS - 304 8 MM
09	TOLERANCE	AS PER DIN - 43760
10	CABLE ENTRY	AS PER MANUFACTURER'S STD.
11	TERMINALS	CERAMIC TERMINAL BLOCK WITH BRASS TERMINAL.
12	THERMOWELL: a) MATERIAL b) CONSTRUCTION c) PROCESS CONNECTION d) RTD CONNECTION e) IMMERSION LENGTH f) TOTAL LENGTH g) ASSEMBLIES	SS - 304 BAR STOCK, STRAIGHT 1 1/2 " 150 # RF SS FLANGE 3/4 " BSP (M) REFER TABLE BELOW - DO - SCREWED TYPE
13	ACCURACY	± 0.5%
14	QUANTITY	4 NOS. (REF. TABLE BELOW)

SL. NO.	TAG NO.	IMMERSION LENGTH	APPROX. TOTAL LENGTH	LOCATION	SYSTEM NO.
01	TE-2207, 2208	22	- DO -	DE OF FAN NO. FN 2200	III
02	TE-2307, 2308	22	REFER DWG INST/01	NDE OF FAN NO. FN 2300	III

RESISTANCE TEMPERATURE DETECTOR WITH THERMOWELL

SL. NO.	DESCRIPTION	
01	MAKE	NAGHAN
02	MATERIAL	PL - 100 WOUND ON CERAMIC
03	RANGE	0 TO 200 DEG.
04	TYPE	3 - WIRE MINERAL, INSULATED / DUPLEX
05	OUTER SHEATH AND DIA	21.3 MM, SS - 304
06	HEAD	DIE CASE AL. EPOXY PAINTED, WEATHER PROOF WITH SS CHAIN & BAKELITE BLOCK & NI PLATED BRASS TERMINALS
07	FILLING	94% Mgo POWDER
08	PROBES: SHEATH OD	SS - 304 8 MM
09	TOLERANCE	AS PER DIN - 43760
10	CABLE ENTRY	AS PER MANUFACTURER'S STANDARD
11	TERMINALS	CERAMIC TERMINAL BLOCK WITH BRASS TERMINAL
12	THERMOWELL: a) MATERIAL b) CONSTRUCTION c) PROCESS CONNECTION d) RTD CONNECTION e) IMMERSION LENGTH f) TOTAL LENGTH g) ASSEMBLIES	SS - 304 BAR STOCK, STRAIGHT 1 1/2 " 150 # RF SS FLANGE REFER DWG. INST/02 REFER TABLE BELOW - DO - FLANGED THERMOWELL *
13	ACCURACY	± 0.5%
14	QUANTITY	1 NOS. (REF. TABLE BELOW)

SLNO	TAG NO.	IMMERSION LENGTH	"U" LENGTH REFER DWG NO. INST/04	LOCATION	SYSTEM NO.
01	TE-2000	1700	1850	3200-G90-N003	III

FLANGED AND CONNECTION SHALL BE AS PER ANSI B 16.5

THERMOCOUPLE WITH THERMOWELL

SL. NO.	DESCRIPTION	
01	MAKE	HAGMAN
02	TYPE	MINERAL INSULATED THERMOCOUPLE SHALL BE OF 14 AWG MAGNESIUM OXIDE INSULATED GROUNDING TYPE.
03	ELEMENT	K-TYPE (CHAI), SIMPLEX
04	STANDARD	AS PER ANSI-MC96-1
05	OUTER SHEATH AND DIA	21.3 MM, SS - 304
06	HEAD	DIE CAST AL. EPOXY PAINTED, WEATHER PROOF WITH SS CHAIN & BAKELITE BLOCK & NI PLATED BRASS TERMINALS
07	OPERATING TEMPERATURE	600 DEG.C
08	CABLE ENTRY	AS PER MANUFACTURER'S STANDARD
09	THERMOWELL: a) MATERIAL b) CONSTRUCTION c) PROCESS CONNECTION d) THERMOCOUPLE CONNECTION	SS - 304 BAR STOCK, STRAIGHT 1 1/2" 150 # RF SS FLANGE REFER DWG INST/02
10	QUANTITY	NOS. (PLEASE REFER TABLE BELOW)

SL.N O.	TAG NO.	IMMERSION LENGTH	"U" LENGTH REFER DWG -INST/04	LOCATION	SYSTEM NO.
01	TE-0250	675	975	1000-G90-S065	III
	TE-0550	675	975	1000-G90-S066	

ELECTRIC TEMPERATURE TRANSMITTER

SL. NO.	DESCRIPTION	
01	MAKE	FISHER - ROSEMOUNT
02	MODEL	0114 TK2-U-1-C-2-NA
03	RANGE, INPUT	K-TYPE (THERMOCOUPLE)/PT-100 (RTD)
04	OUTPUT	2-WIRE, LINEAR WITH MILLIVOLT INPUT SIGNAL FOR THERMOCOUPLE/RTD INPUT FOR PT - 100
05	ACCURACY	± 0.2% OF CALIBRATED SPAN
06	STABILITY	± 0.2% OF CALIBRATED SPAN FOR 6 MONTHS
07	AMBIENT TEMPERATURE LIMIT	- 25 DEG.C TO + 80 DEG.C
08	POWER SUPPLY	24 VOLT DC
09	LOSS OF IP INDICATION	UPSCALE BURN-OUT PROTECTION PROVIDED
10	LOCAL INDICATION	TO BE PROVIDED

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IP

(BCH)

SL.NO.	TAG NO.	INPUT	RANGE (Deg. C)	SYSTEM NO.	QUANTITY
01	TT-0250	K-TYPE THERMOCOUPLE	0-600	III	1
02	TT-0550	K-TYPE THERMOCOUPLE	0-600	III	1
03	TT-2000	PT-100 RTD	0-200	III	1

TECHNICAL SPECIFICATION FOR DUST LEVEL SWITCH

SL. NO.	DESCRIPTION	
01	TAG NO.	DLS-2001 TO DLS - 2012 (SYSTEM - III)
02	QUANTITY	12 (SYS.III)
03	AMBIENT TEMPERATURE	50 DEG. C
04	PROCESS TEMPERATURE	55 DEG. C
05	PROCESS PRESSURE	700 mm WG
06	TYPE	RADIO FREQUENCY
07 *	PROBE a) LENGTH b) MATERIAL c) INSULATION d) PROCESS CONNECTION	750 MM OVERALL INCLUDING PROBE/ EXTENSION 33-304 TEFLON FLANGE
08 *	<u>ELECTRONIC CONTROLLER</u>	
09	POWER SUPPLY	240V AC $\pm$ 10%, 50 Hz $\pm$ 3% #
10	ALARM	POTENTIAL FREE, 2 NOS. (1 NO + 1 NC) RATED AT 3 AMP. 24V DC
11	LOCAL LED INDICATIONS	NORMAL LEVEL ALARM LEVEL PROBE HEALTHY
12	SPECIAL CABLE	10 MTRS. SHALL BE SUPPLIED

* PROBE AND ELECTRONIC CONTROLLER WERE SEPARATELY MOUNTED.
PS WILL BE TAPPED FROM MCC OF INDIVIDUAL SYSTEM.

TECHNICAL SPECIFICATION FOR DIFFERENTIAL PRESSURE SWITCH

SL. NO.	DESCRIPTION	
01	TAG NO.	DFS-2000
02	QUANTITY	1 NOS.
03	RANGE	25 TO 250 MM WG
04	PROCESS	DUSTY AIR
05	MAX. TEMPERATURE OF THE PROCESS	150 DEG.C
06	MAX. STATIC PRESSURE OF THE SYSTEM	0.5 BAR
07	AMBIENT TEMPERATURE	50 DEG.C
08	MEASURING ELEMENT	DIAPHRAGM
09	MATERIAL	NEOPRENE
10	PROCESS CONNECTION	1/2" NPTF AS PER MANUFACTURER STD.
11	ELECTRICAL	1/2" NPTF AS PER MANUFACTURER STD.
12	ENCLOSURE	IP-55
13	ACCURACY	± 2%
14	<u>SWITCH UNIT</u>	
	- OUTPUT	MICROSWITCH ENVIRONMENTALLY SEALED. TWO INDEPENDENT ADJUSTABLE POT FREE SET POINTS EACH WITH 1 NO + 1 NC, 3 A CONTACT FOR DIFFERENT SETTING RANGE. SUITABLE FOR 240 V SYSTEM.
	- SWITCH SETTING	FIELD ADJUSTABLE OVER FULL RANGE
	- DIFFERENTIAL	FIXED (≤ 5% FSR)

TECHNICAL SPECIFICATION FOR PRESSURE SWITCH

SL. NO.	DESCRIPTION	
01	TAG NO.	PS - 2000
02	QUANTITY	1 NOS.
03	RANGE	1 - 7.5 KG/CM ²
04	PROCESS	COMPRESSED AIR
05	MAX. TEMPERATURE OF PROCESS	55 DEG. C
06	MEASURING ELEMENT AND MATERIAL	PHOSPHOR BRONZE BELLOWS
07	PROCESS CONNECTION	1/2" NPTF
08	ELECTRICAL CONNECTION	3/4" ET CABLE GLAND
09	ENCLOSURE	IP - 65
10	SWITCH UNIT	
	- OUTPUT	MICROSWITCH ENVIRONMENTALLY SEALED CONTACT. TYPE / RATING SPDT / 3 Amp. 24 V D.C (1 NO + 1 NC)
	- SET POINT	SETTABLE OVER FULL RANGE WITH LOCK AND OVER TRAVEL STOP.

TECHNICAL SPECIFICATION FOR PRESSURE TRANSMITTER

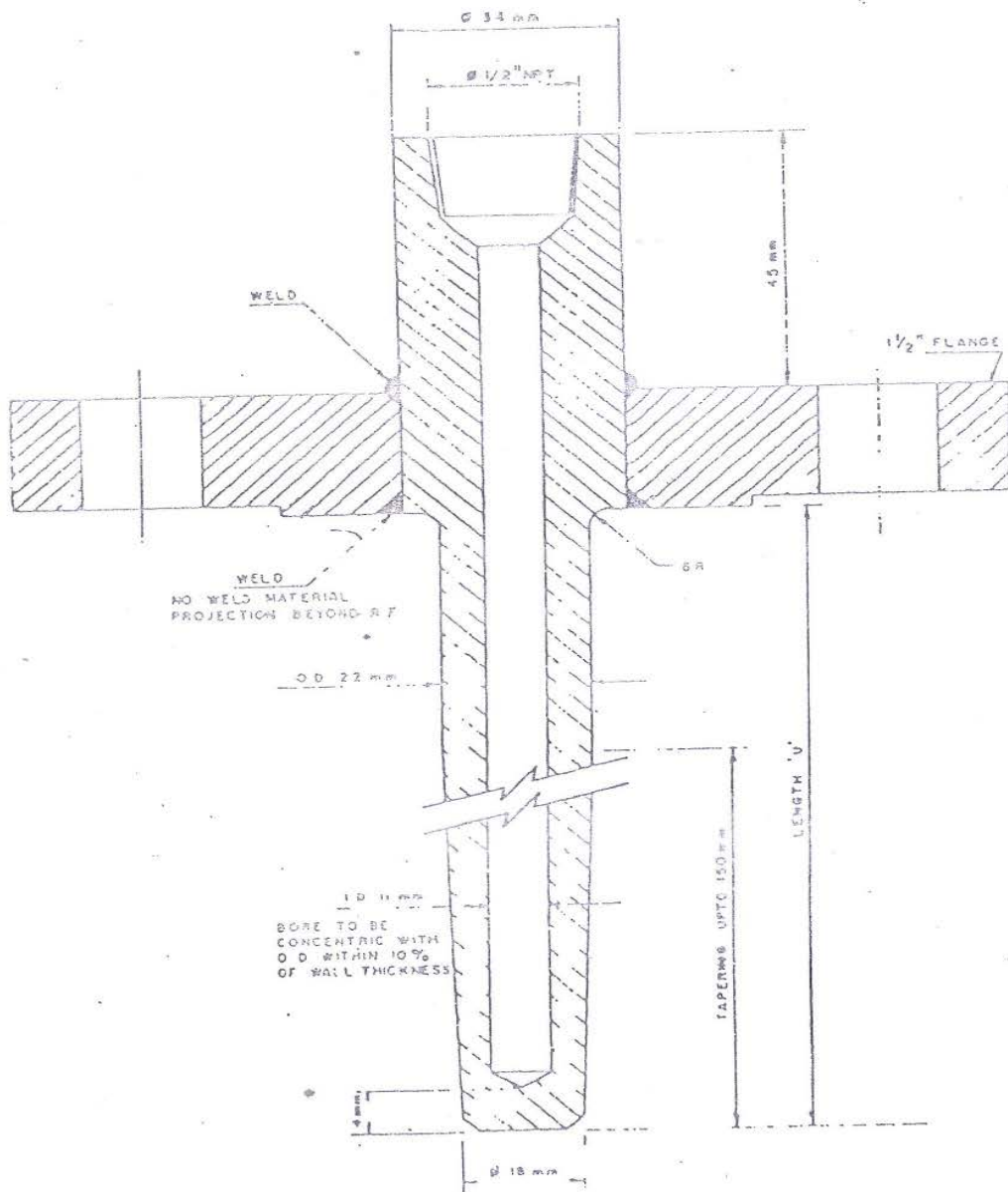
SL. NO.	DESCRIPTION	
01	FUNCTION	TO MEASURE GAUGE PRESSURE
02	SENSOR	CAPACITANCE
03	MAKE / MODEL	FISHER ROSEMOUNT / H181DR2F42A1N1
04	POWER	24 V DC
05	OUTPUT	4 - 20 mA DC (TWO WIRE)
06	OUTPUT INDICATION	0 - 100% LINEAR
07	<u>ELECTRONIC OPERATING</u>	
	- AMBIENT TEMP. LIMIT	(-) 40 TO 93 DEG. C
	- HUMIDITY	0 - 100% RH
	- PROCESS CONNECTION	½ " NPTF WITH FLANGE ADAPTER
	- BODY	CADMIUM PLATED CARBON STEEL
	- ELECTRICAL CONNECTION	½ " NPTF
08	TAG NO. & QTY.	AS TABULATED BELOW
09	ACCESSORIES	3 WAY VALVE MANIFOLD
10	ACCURACY	±0.075% of FS

SL.NO.	TAG NO.	RANGE	SYSTEM NO.	QUANTITY
01	PT-0250	-100 TO 0 Pa	III	1
02	PT-0550	-100 TO 0 Pa	III	1
03	PT-2000	-5 TO 0 KPa	III	1

TECHNICAL SPECIFICATION FOR PNEUMATIC PANEL

SL. NO.	DESCRIPTION	
01	PANEL MATERIAL	2 MM CRCA STEEL
02	PAINT OUTSIDE PAINT INSIDE	EPOXY PAINTED AS PER SHADE 631 OF IS - 5 EPOXY GLOSSY WHITE
03	ENCLOSURE	IP-55 WITH CANOPY
04	NIPPLE	SERRATED BRASS NIPPLE WITH HOSE CLAMP
05	TERMINAL	2.5 SQ. MM PVC MATERIAL ELMAX MAKE
06	ACCESSORIES	20 NB FILTER REGULATED CUB LUBRICATOR UNIT, 20 NB BALL VALVE SS 316 TUBING AS PER ENCLOSED SKETCH.
07	SOLENOID VALVE	1/4 " BSP
08	TYPE	4 WAY 5 PORT SINGLE COIL CONTINUOUS DUTY
09	STYLE OF COIL	EPOXY MOULDED, CLASS F
10	VOLTAGE OF COIL	24 V, DC, 1 PHASE
11	MANUAL OVERRIDE FACILITY	REQUIRED
12	ENCLOSURE	IP - 67
13	TEMPERATURE	50 DEG.C
14	QTY.	TYPE - A : 2 NOS. (2 NOS. 5 PORT/4 WAY SOLENOID VALVE & FILTER REGULATOR UNIT WITH FITTING) TYPE - B : 5 NOS. (3 NOS. 5 PORT/ 4 WAY SOLENOID VALVE AND FILTER REGULATOR UNIT WITH FITTING)
15	TAG NO.	AS DETAILED IN DRG. NO. INST/03
16	LOCATION	REFER ABOVE DRAWINGS

INSTRUMENTATION

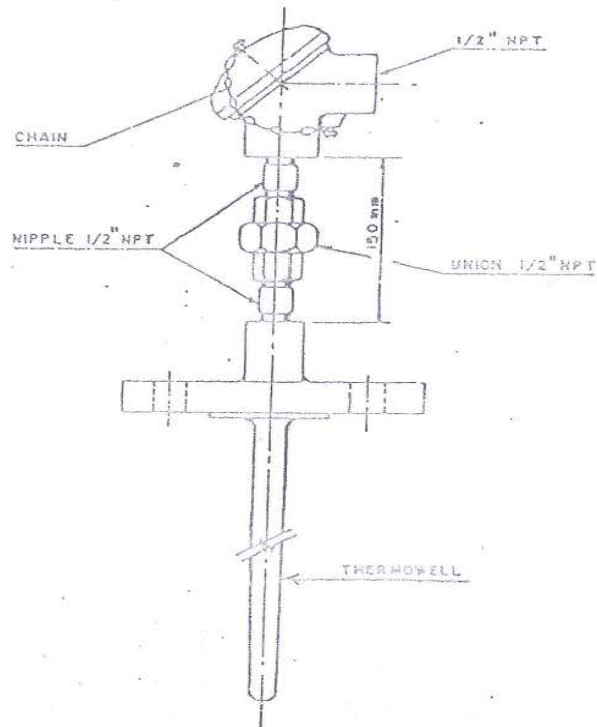


NOTE THIS STANDARD IS APPLICABLE
 1) FOR TEMPERATURE GAUGES AND BEADED THERMOCOUPLES
 2) UPTO 600° ANSI OR EQUIVALENT RATINGS

THERMOWELL

DWG NO. - INST/04

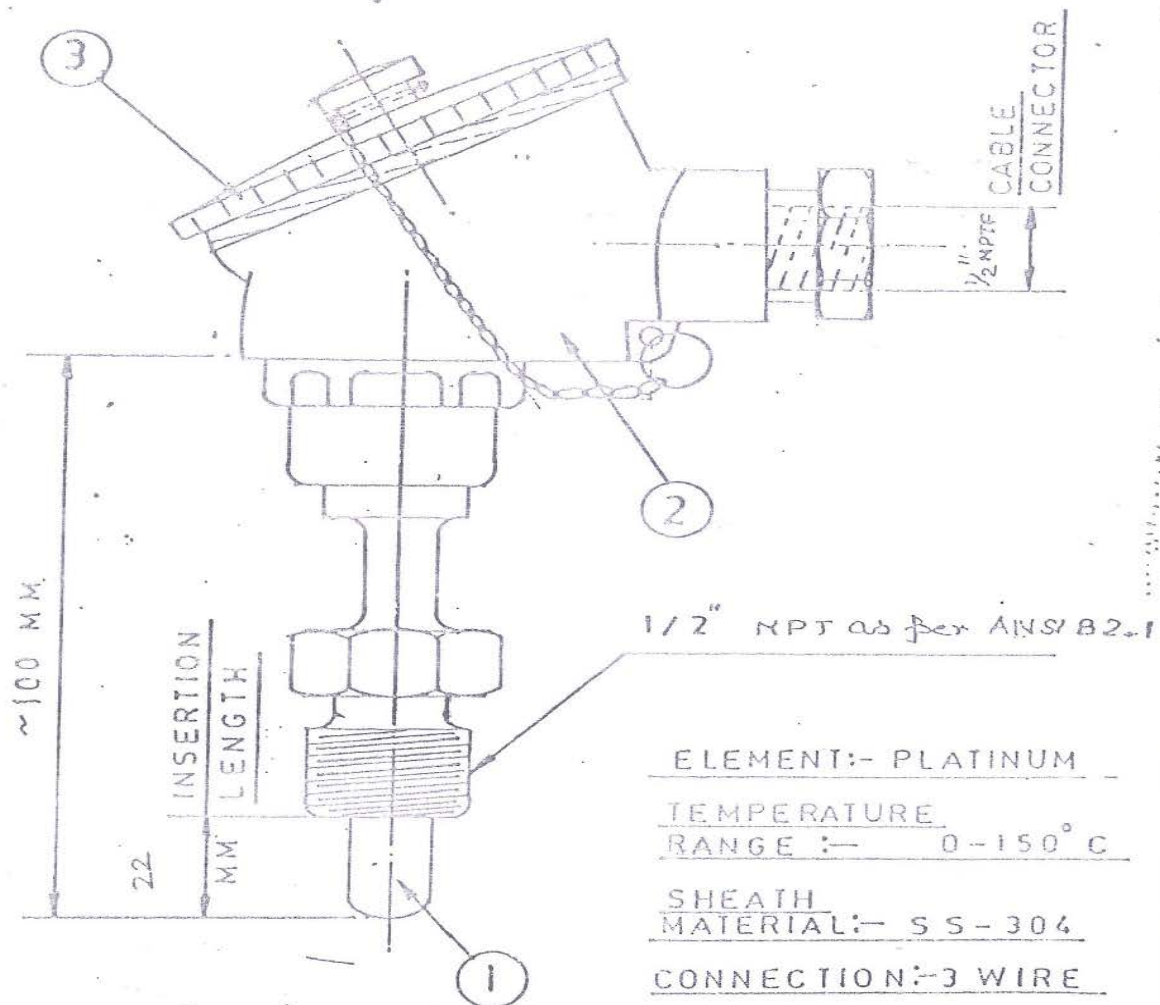
INSTRUMENTATION



NOTE IN THE CASE OF DUPLEX THERMOCOUPLE / RTD, TWO INDEPENDENT ELECTRICAL ENTRIES SHALL BE PROVIDED

THERMOCOUPLE / RTD
ASSEMBLY WITH THERMOWELL

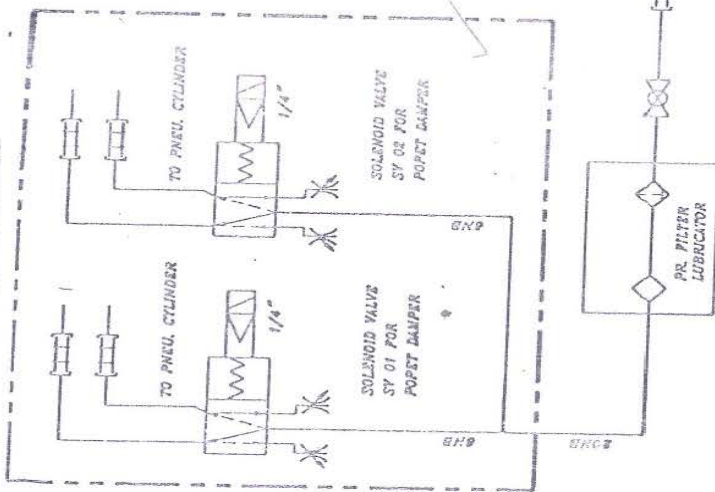
DWG NO. - INST/02



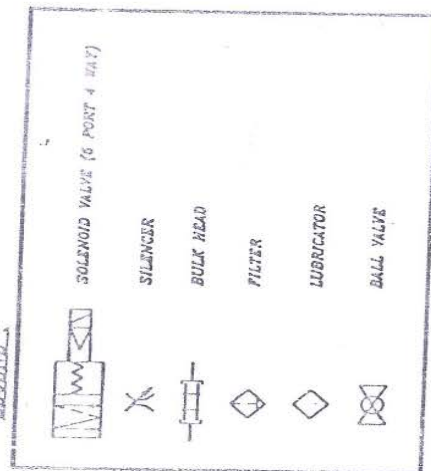
1	3	CAP WITH CHAIN COVER	THREADED	ALUMINIUM
1	2	COVER	THREADED	CAST ALUMINIUM
1	1	ELEMENT		PLATINUM

DWG NO. - INST/01

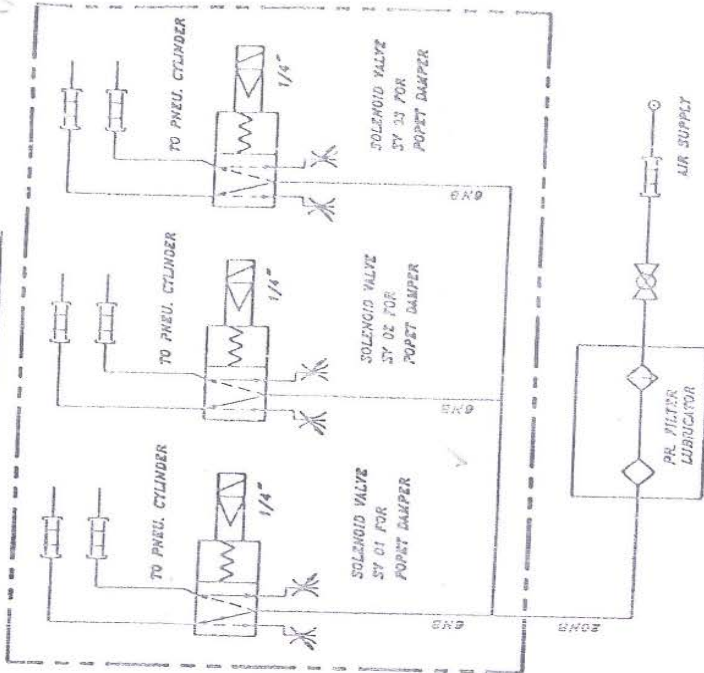
TYPE - A



LEGEND:



TYPE - B



SL. NO	TAC NO.	TYPE	QTY	SYSTEM
1	PCB-2000 A1&A2. 2000 B1&B2	B	4	III

NOTE: FOR LOCATION OF PCB, REF. DRG. NO.

INDICATES INSTRUMENTS INSI.

PROJECT : GUJARAT COPPER PROJECT

CABLE SCHEDULE OF
INSTRUMENTATION & AUTOMATION
FOR DEDUSTING SYSTEM - III

Doc. No. HCL/GCP/12-III/INST-01 Rev 0

No. of Sheets : 8

					CABLE DESCRIPTION				
SL. NO.	CABLE TAG NO.	FROM	THRU	TO	NO. OF CORES	SIZE (MM ²)	TYPE	VOLTS	LENGTH (M)
	SYSTEM - III								
01	01-JS-001	JUNCTION BOX 03-JBS-001	TRAY	PRESSURE TRANSMITTER PT-2000	3/C	1.5	YVY (SCREEN)	1100	10
02	01-JS-002	JUNCTION BOX 03-JBS-001	TRAY	PLC-3	3/C	1.5	YVY (SCREEN)	1100	50
03	02-JS-001	JUNCTION BOX 03-JBS-002	TRAY	DUST LEVEL MONITOR DLS - 2001	3/C	1.5	YVY	1100	15
04	02-JS-002	JUNCTION BOX 03-JBS-002	TRAY	DUST LEVEL MONITOR DLS - 2002		1.5	YVY	1100	10
05	02-JS-003	JUNCTION BOX 03-JBS-002	TRAY	DUST LEVEL MONITOR DLS - 2003	3/C	1.5	YVY	1100	15
06	02-JS-004	JUNCTION BOX 03-JBS-002	TRAY	DUST LEVEL MONITOR DLS - 2004	3/C	1.5	YVY	1100	15
07	02-JS-005	JUNCTION BOX 03-JBS-002	TRAY	DUST LEVEL MONITOR DLS - 2005	3/C	1.5	YVY	1100	20
08	02-JS-006	JUNCTION BOX 03-JBS-002	TRAY	DUST LEVEL MONITOR DLS - 2006	3/C	1.5	YVY	1100	20
09	02-JS-007	JUNCTION BOX 03-JBS-002	TRAY	INLET DAMPER (LTSW) ID-2001	3/C	1.5	YVY	1100	20
10	02-JS-008	JUNCTION BOX 03-JBS-002	TRAY	INLET DAMPER (LTSW) ID-2002	3/C	1.5	YVY	1100	25
11	02-JS-009	JUNCTION BOX 03-JBS-002	TRAY	INLET DAMPER (LTSW) ID-2003	3/C	1.5	YVY	1100	25
12	02-JS-010	JUNCTION BOX 03-JBS-002	TRAY	INLET DAMPER (LTSW) ID-2004		1.5	YVY	1100	20
13	02-JS-011	JUNCTION BOX 03-JBS-002	TRAY	INLET DAMPER (LTSW) ID-2005	3/C	1.5	YVY	1100	20
14	02-JS-012	JUNCTION BOX 03-JBS-002	TRAY	INLET DAMPER (LTSW) ID-2006	3/C	1.5	YVY	1100	25
15	02-JS-013	JUNCTION BOX 03-JBS-002	TRAY	PLC	10/C	1.5	YVY	1100	50
16	03-JS-001	JUNCTION BOX 03-JBS-003	TRAY	PNEUMATIC PANEL PCB-2000A1A2	10/C	1.5	YVY	1100	15 + 15 = 30

SL NO.	CABLE TAG NO.	FROM	THRU	TO	CABLE DESCRIPTION				
					NO. OF CORES	SIZE (MM ²)	TYPE	VOLTS	LENGTH (M)
17	03-JS-002	JUNCTION BOX 03-JBS-003	TRAY	PLC	10/C	1.5	YVY	1100	75
18	04-JS-001	JUNCTION BOX 03-JBS-004	TRAY	PNEUMATIC PANEL PCB-2000D1,B2	10/C	1.5	YVY	1100	15 + 15 = 30
19	04-JS-002	JUNCTION BOX 03-JBS-004	TRAY	PLC	10/C	1.5	YVY	1100	75
20	05-JS-001	JUNCTION BOX 03-JBS-005	TRAY	LT. SW. FOR POPPET DAMPER LSO-2000A1	3/C	1.5	YVY	1100	10
21	05-JS-002	JUNCTION BOX 03-JBS-005	TRAY	LT. SW. FOR POPPET DAMPER LSO-2000A2	3/C	1.5	YVY	1100	10
22	05-JS-003	JUNCTION BOX 03-JBS-005	TRAY	LT. SW. FOR POPPET DAMPER LSO-2000A3	3/C	1.5	YVY	1100	15
23	05-JS-004	JUNCTION BOX 03-JBS-005	TRAY	LT. SW. FOR POPPET DAMPER LSO-2000A4	3/C	1.5	YVY	1100	15
24	05-JS-005	JUNCTION BOX 03-JBS-005	TRAY	LT. SW. FOR POPPET DAMPER LSO-2000A5	3/C	1.5	YVY	1100	20
25	05-JS-006	JUNCTION BOX 03-JBS-005	TRAY	LT. SW. FOR POPPET DAMPER LSO-2000A6	3/C	1.5	YVY	1100	20
26	05-JS-007	JUNCTION BOX 03-JBS-005	TRAY	LT. SW. FOR POPPET DAMPER LSC-2000A1	3/C	1.5	YVY	1100	15
27	05-JS-008	JUNCTION BOX 03-JBS-005	TRAY	LT. SW. FOR POPPET DAMPER LSC-2000A2	3/C	1.5	YVY	1100	15
28	05-JS-009	JUNCTION BOX 03-JBS-005	TRAY	LT. SW. FOR POPPET DAMPER LSC-2000A3	3/C	1.5	YVY	1100	20
29	05-JS-010	JUNCTION BOX 03-JBS-005	TRAY	LT. SW. FOR POPPET DAMPER LSC-2000A4	3/C	1.5	YVY	1100	20
30	05-JS-011	JUNCTION BOX 03-JBS-005	TRAY	LT. SW. FOR POPPET DAMPER LSC-2000A5	3/C	1.5	YVY	1100	10
31	05-JS-012	JUNCTION BOX 03-JBS-005	TRAY	LT. SW. FOR POPPET DAMPER LSC-2000A6	3/C	1.5	YVY	1100	10
32	05-JS-013	JUNCTION BOX 03-JBS-005	TRAY	PLC	10/C	1.5	YVY	1100	75

SL. NO.	CABLE TAG NO.	FROM	THRU	TO	CABLE DESCRIPTION				
					NO. OF CORES	SIZE (MM ²)	TYPE	VOLTS	LENGTH (M)
33	06-JS-001	JUNCTION BOX 03-JBS-006	TRAY	DIFFERENTIAL PRESSURE SWITCH DPS - 2000	5/C	1.5	YVY	1100	75
34	06-JS-002	JUNCTION BOX 03-JBS-006	TRAY	54 NOS. SOLENOID VALVE	54 X 3/C	1.5	YVY	1100	54 X 10 = 540
35	06-JS-003	JUNCTION BOX 03-JBS-006	TRAY	PRESSURE SWITCH PSL-2000	3/C	1.5	YVY	1100	15
36	06-JS-004	JUNCTION BOX 03-JBS-006	TRAY	PLC	19/C	1.5	YVY	1100	75
37	07-JS-001	JUNCTION BOX 03-JBS-007	TRAY	54 NOS. SOLENOID VALVE	54 X 3/C	1.5	YVY	1100	54 X 10 = 540
38	07-JS-002	JUNCTION BOX 03-JBS-007	TRAY	PLC	19/C	1.5	YVY	1100	75
39	08-JS-001	JUNCTION BOX 03-JBS-008	TRAY	DUST LEVEL MONITOR DLS - 2007	3/C	1.5	YVY	1100	10
40	08-JS-002	JUNCTION BOX 03-JBS-008	TRAY	DUST LEVEL MONITOR DLS - 2008	3/C	1.5	YVY	1100	10
41	08-JS-003	JUNCTION BOX 03-JBS-008	TRAY	DUST LEVEL MONITOR DLS - 2009	3/C	1.5	YVY	1100	15
42	08-JS-004	JUNCTION BOX 03-JBS-008	TRAY	DUST LEVEL MONITOR DLS - 2010	3/C	1.5	YVY	1100	15
43	08-JS-005	JUNCTION BOX 03-JBS-008	TRAY	DUST LEVEL MONITOR DLS - 2011	3/C	1.5	YVY	1100	20
44	08-JS-006	JUNCTION BOX 03-JBS-008	TRAY	DUST LEVEL MONITOR DLS - 2012	3/C	1.5	YVY	1100	20
45	08-JS-007	JUNCTION BOX 03-JBS-008	TRAY	INLET DAMPER (LTSW) ID - 2007	3/C	1.5	YVY	1100	15
46	08-JS-008	JUNCTION BOX 03-JBS-008	TRAY	INLET DAMPER (LTSW) ID - 2008	3/C	1.5	YVY	1100	15
47	08-JS-009	JUNCTION BOX 03-JBS-008	TRAY	INLET DAMPER (LTSW) ID - 2009	3/C	1.5	YVY	1100	15
48	08-JS-010	JUNCTION BOX 03-JBS-008	TRAY	INLET DAMPER (LTSW) ID - 2010	3/C	1.5	YVY	1100	15

SL. NO.	CABLE TAG NO.	FROM	THRU	TO	CABLE DESCRIPTION				
					NO. OF CORES	SIZE (MM ²)	TYPE	VOLTS	LENGTH (M)
49	08-JS-011	JUNCTION BOX 03-JBS-008	TRAY	INLET DAMPER (LTSW) ID - 2011	3/C	1.5	YVY	1100	15
50	08-JS-012	JUNCTION BOX 03-JBS-008	TRAY	INLET DAMPER (LTSW) ID - 2012	3/C	1.5	YVY	1100	15
51	08-JS-013	JUNCTION BOX 03-JBS-008	TRAY	PLC	19/C	1.5	YVY	1100	20
52	09-JS-001	JUNCTION BOX 03-JBS-009	TRAY	LT. SW. FOR POPPET DAMPER LSO-2000B1	3/C	1.5	YVY	1100	10
53	09-JS-002	JUNCTION BOX 03-JBS-009	TRAY	LT. SW. FOR POPPET DAMPER LSO-2000B2	3/C	1.5	YVY	1100	10
54	09-JS-003	JUNCTION BOX 03-JBS-009	TRAY	LT. SW. FOR POPPET DAMPER LSO-2000B3	3/C	1.5	YVY	1100	15
55	09-JS-004	JUNCTION BOX 03-JBS-009	TRAY	LT. SW. FOR POPPET DAMPER LSO-2000B4	3/C	1.5	YVY	1100	15
56	09-JS-005	JUNCTION BOX 03-JBS-009	TRAY	LT. SW. FOR POPPET DAMPER LSO-2000B5	3/C	1.5	YVY	1100	20
57	09-JS-006	JUNCTION BOX 03-JBS-009	TRAY	LT. SW. FOR POPPET DAMPER LSO-2000B6	3/C	1.5	YVY	1100	20
58	09-JS-007	JUNCTION BOX 03-JBS-009	TRAY	LT. SW. FOR POPPET DAMPER LSC-2000B1	3/C	1.5	YVY	1100	20
59	09-JS-008	JUNCTION BOX 03-JBS-009	TRAY	LT. SW. FOR POPPET DAMPER LSC-2000B2	3/C	1.5	YVY	1100	20
60	09-JS-009	JUNCTION BOX 03-JBS-009	TRAY	LT. SW. FOR POPPET DAMPER LSC-2000B3	3/C	1.5	YVY	1100	15
61	09-JS-010	JUNCTION BOX 03-JBS-009	TRAY	LT. SW. FOR POPPET DAMPER LSC-2000B4	3/C	1.5	YVY	1100	15
62	09-JS-011	JUNCTION BOX 03-JBS-009	TRAY	LT. SW. FOR POPPET DAMPER LSC-2000B5	3/C	1.5	YVY	1100	10
63	09-JS-012	JUNCTION BOX 03-JBS-009	TRAY	LT. SW. FOR POPPET DAMPER LSC-2000B6	3/C	1.5	YVY	1100	10
64	09-JS-013	JUNCTION BOX 03-JBS-009	TRAY	PLC	19/C	1.5	YVY	1100	25

SL. NO.	CABLE TAG NO.	FROM	THRU	TO	CABLE DESCRIPTION				
					NO. OF CORES	SIZE (MM ²)	TYPE	VOLTS	LENGTH (M)
65	10-JS-001	JUNCTION BOX 03-JBS-010	TRAY	54 NOS SOLENOID VALVE	54 X 3/C	1.5	YWW	1100	54 X 10 = 550
66	10-JS-002	JUNCTION BOX 03-JBS-010	TRAY	PLC	19/C	1.5	YWW	1100	75
67	11-JS-001	JUNCTION BOX 03-JBS-011	TRAY	54 NOS. SOLENOID VALVE	54 X 3/C	1.5	YWW	1100	54 X 10 = 550
68	11-JS-002	JUNCTION BOX 03-JBS-011	TRAY	PLC	19/C	1.5	YWW	1100	75
69	12-JS-001	JUNCTION BOX 03-JBS-012	TRAY	TEMPERATURE ELEMENT (RTD) TE - 2000	3/C	1.5	YWW	1100	15
70	12-JS-002	JUNCTION BOX 03-JBS-012	TRAY	TEMPERATURE TRANSMITTER TT - 2000	3/C	1.5	YWW (SCREEN)	1100	15
71	12-JS-003	JUNCTION BOX 03-JBS-012	TRAY	PLC - 3	3/C	1.5	YWW	1100	60
72	12-JS-004	JUNCTION BOX 03-JBS-012	TRAY	ELECTRO PNEUMATIC DAMPER TV - 2000	3/C	1.5	YWW (SCREEN)	1100	50
73	12-JS-005	JUNCTION BOX 03-JBS-012	TRAY	POS. TR. OF ACTUATOR ZT-2000	3/C	1.5	YWW (SCREEN)	1100	50
74	12-JS-006	JUNCTION BOX 03-JBS-012	TRAY	PLC - 3	5/C	1.5	YWW (SCREEN)	1100	60
75	13-JS-001	JUNCTION BOX 03-JBS-013	TRAY	THERMOCOUPLE TE-250 (K-TYPE)	3/C	1.5	YWW COMPEN- SETTING CABLE	1100	15
76	14-JS-001	JUNCTION BOX 03-JBS-014	TRAY	THERMOCOUPLE TE-550 (K-TYPE)	3/C	1.5	YWW COMPEN- SETTING CABLE	1100	15
77	15-JS-001	JUNCTION BOX 03-JBS-015	TRAY	ACTUATOR XVM - 2200	3/C	1.5	YWW (SCREEN)	1100	10
78	15-JS-002	JUNCTION BOX 03-JBS-015	TRAY	PLC	3/C	1.5	YWW (SCREEN)	1100	40
79	15-JS-003	JUNCTION BOX 03-JBS-015	TRAY	ELECTROHYD POSITIONER CU-01 OF ACTUATOR	7/C	1.5	YWW	1100	15

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SL. NO.	CABLE TAG NO.	FROM	THRU	TO	CABLE DESCRIPTION				
					NO. OF CORES	SIZE (MM ²)	TYPE	VOLTS	LENGTH (M)
80	15-JS-004	JUNCTION BOX 03-JBS-015	TRAY	PLC	5/C	1.5	YVY	1100	40
81	15-JS-005	JUNCTION BOX 03-JBS-015	TRAY	ACTUATOR XVM - 2300	3/C	1.5	YVY (SCREEN)	1100	15
82	15-JS-006	JUNCTION BOX 03-JBS-015	TRAY	PLC	3/C	1.5	YVY (SCREEN)	1100	50
83	15-JS-007	JUNCTION BOX 03-JBS-015	TRAY	ELECTRONICS POSITIONER CU-01 OF ACTUATOR	7/C	1.5	YVY	1100	20
84	15-JS-008	JUNCTION BOX 03-JBS-015	TRAY	PLC	5/C	1.5	YVY	1100	50
85	15-JS-009	JUNCTION BOX 03-JBS-015	TRAY	PNEUMATIC PANEL PCB-2200	5/C	1.5	YVY	1100	15
86	16-JP-001	JUNCTION BOX 03-JBP-016	TRAY	MCC - 3 (IA3T453)	3/C	4	AYVY	1100	60
87	16-JP-002	JUNCTION BOX 03-JBP-016	TRAY	DUST LEVEL MONITOR DLM - 2001	3/C	1.5	YVY	1100	10
88	16-JP-003	JUNCTION BOX 03-JBP-016	TRAY	DUST LEVEL MONITOR DLM - 2002	3/C	1.5	YVY	1100	10
89	16-JP-004	JUNCTION BOX 03-JBP-016	TRAY	DUST LEVEL MONITOR DLM - 2003	3/C	1.5	YVY	1100	15
90	16-JP-005	JUNCTION BOX 03-JBP-016	TRAY	DUST LEVEL MONITOR DLM - 2004	3/C	1.5	YVY	1100	15
91	16-JP-006	JUNCTION BOX 03-JBP-016	TRAY	DUST LEVEL MONITOR DLM - 2005	3/C	1.5	YVY	1100	15
92	16-JP-007	JUNCTION BOX 03-JBP-016	TRAY	DUST LEVEL MONITOR DLM - 2006	3/C	1.5	YVY	1100	15
93	16-JP-008	JUNCTION BOX 03-JBP-016	TRAY	DUST LEVEL MONITOR DLM - 2007	3/C	1.5	YVY	1100	20
94	16-JP-009	JUNCTION BOX 03-JBP-016	TRAY	DUST LEVEL MONITOR DLM - 2008	3/C	1.5	YVY	1100	20
95	16-JP-010	JUNCTION BOX 03-JBP-016	TRAY	DUST LEVEL MONITOR DLM - 2009	3/C	1.5	YVY	1100	10

SL. NO.	CABLE TAG NO.	FROM	THRU	TO	CABLE DESCRIPTION				
					NO. OF CORES	SIZE (MM ²)	TYPE	VOLTS	LENGTH (M)
96	16-JP-011	JUNCTION BOX 03-JBP-018	TRAY	DUST LEVEL MONITOR DLM - 2010	3/C	1.5	YVY	1100	
97	17-JP-001	MCC - 3	TRAY	PNEUMATIC PANEL PCB - 2000A1	3/C	1.5	YVY	1100	
98	17-JP-002	MCC - 3	TRAY	PNEUMATIC PANEL PCB - 2000B1	3/C	1.5	YVY	1100	
99	17-JP-003	MCC - 3	TRAY	PNEUMATIC PANEL PCB - 2000A2	3/C	1.5	YVY	1100	
100	17-JP-004	MCC - 2	TRAY	PNEUMATIC PANEL PCB - 2000B2	3/C	1.5	YVY	1100	

ELECTRICAL CODES & STANDARDS

The following Codes and Standards shall be applicable for Electrical Facilities.

Sr. No.	Codes & Standards	Description
1	IS-5	Colors for Ready Mixed Paints and Enamels
2	IS-335	New Insulating Oils.
3	IS-2071 (P1 to P3)	Methods of High Voltage Testing.
4	IS-12063	Classification of degrees of protection provided by enclosures of electrical equipment.
5	IS-2165 P1:1997 P2:1983	Insulation Coordination
6	IS-1448 (P1 to P145)	Methods of Test for Petroleum and its Products.
7	IS-6792	Method for determination of electric strength of insulating oils
8	IS-5578	Guide for marking of insulated conductors
9	IS-11353	Guide for uniform system of marking & identification of conductors & apparatus terminals
10	IS: 10028 (Part 2&3), (P1:1993), (P2:1991), (P3:1991)	Code of practice for selection, installation & maintenance of Transformers
11	IS-2026 (P1 to P4)	Power Transformers
12	IS-3639	Fittings and Accessories for Power Transformers
13	IS-2705- (P1 to P4)	Current Transformers
14	IS-3156- (P1 to P4)	Voltage Transformers
15	IEC-56	High Voltage Alternating Current Circuit Breakers
16	IS-6600	Guide for Loading of Oil immersed Transformers.
17	IS-10118	Code of Practice for selection, installation & maintenance of Switchgear & Control Gear.
18	IS-7689	Guide for Control of undesirable static electricity.
19	IS-5216	Guide for safety procedures and practices in electrical work.
20	IS 9676	Reference ambient temperature for electrical equipment.
21	IS 3043	Code of practice for Earthing
22	IS 2309	Code of Practice for Protection of Buildings & Allied Structures against Lightning.
23	IS-722, IS-1248, IS-3231 (P-3)	Electrical relays for power system protection.
24	IS-1448 (P1 to P145)	Methods of Test for Petroleum and its Products.
25	IS-6103	Method of Test for Specific Resistance (Resistivity) of Electrical Insulating Liquids
26	IS-6104	Method of Test for Interfacial Tension of Oil against Water by the Ring Method
27	IS-6262	Method of test for Power factor & Dielectric Constant of Electrical Insulating Liquids

Sr. No.	Codes & Standards	Description
28	IS-9224 (Part 1,2&4)	Low Voltage Fuses
29	IS-731	Porcelain insulators for overhead power lines with a nominal voltage greater than 1000 V
30	IS-2486 (P1 to P4)	Insulator fittings for overhead power lines with a nominal voltage greater than 1000 V
31	IS-694	PVC insulated cables for working voltages up to and including 1100 Volts
32	IS-1255	Code of practice for installation and maintenance of power cables, up to and including 33 kV rating
33	IS-1554 (P1 and P2)	PVC insulated (heavy duty) electric cables (part 1)for working voltage up to and including 1100 V.- Part (2) for working voltage from 3.3 kV up to and including 11Kv
34	IS: 1753	Aluminum conductor for insulated cables
35	IS: 2982	Copper Conductor in insulated cables
36	IS-3961 (P1 to P5)	Recommended current ratings for cables
37	IS-3975	Mild steel wires, formed wires and tapes for armouring of cables
38	IS-5831	PVC insulating and sheath of electric cables
39	IS-7098	Cross-linked polyethylene insulated PVC sheathed cables for working voltage from 3.3kV up to and including 33 kV
40	IS-8130	Conductors for insulated electrical cables and flexible cords
41	IS-10418	Specification for drums for electric cables
42	IS-2629	Recommended Practice for Hot-Dip galvanizing on iron and steel
43	IS-2633	Methods for testing uniformity of coating of zinc coated articles
44	IS-6005	Code of practice for phosphating of iron and steel.
45	SSPEC	Steel structure painting council
46	IS: 4064	Air break switches
47	IS: 9224(Part-II)	HRC Fuses
48	IS: 6875	Push Button and Control Switches
49	IS: 1248	Indicating instruments
50	IS: 375	Auxiliary wiring & busbar markings
51	IS: 2959	Contactors
52	IS: 1521 ISO/R: 89-1959	Method for Tensile Testing of Steel Wire
53	IS: 1913	General and safety requirements for electric lighting fittings
54	IS: 4012	Dust proof electric lighting fittings
55	IS: 4013	Dust tight proof electric lighting fittings
56	IS: 10322	Industrial lighting fittings with metal reflectors
57	IS: 10322	Industrial lighting fittings with plastic reflectors
58	IS: 10322	Luminaries for street lighting
59	IS: 9900	High pressure mercury vapour lamps
60	IS: 9537	Rigid steel conduits for electrical wiring
61	IS: 3480	Flexible steel conduits for electrical wiring
62	IS: 13947	LV Switchgear and Control gear (Part 1 to 5)
63	IS: 2551	Danger notice plates
64	IS: 8623	Factory built Assemblies of Switchgear and Control Gear for voltages up to

Sr. No.	Codes & Standards	Description
		and including 1000V AC and 1200V DC
65	IS: 1293	3 pin plug
66	IS: 3854	Switches for domestic and similar purposes
67	IS: 5216	Guide for safety procedures and practices in electrical work.
68	IS: 732	Code of practice for electrical wiring installation (system voltage not exceeding 650 Volts.)
69	IS: 3646	Code of practice of interior illumination part II & III
70	IS: 13134	Guide for selection of insulators in respect of polluted atmosphere
71	IS: 325 : 1996	Three phase induction motors (fifth revision)
72	IS: 4029 : 2010	Guide for testing three phase induction motors
73	IS: 4722 : 2001	Rotating electrical machines – Specification
74	IS: 4889 : 1968	Methods of determination of efficiency of rotating electrical machines
75	IS: 7816 : 1975	Guide for testing insulation resistance of rotating machines
76	IS: 12065: 1987	Permissible limits of noise level for rotating electrical machines

Media Code for the Project as per Process Licensor

Sr. No.	Content	Media Code
1	Drinking Water	-
2	De-Mineralized Water	W 10
3	Condensate	C 10
4	Condensate, Oxide	C 40
5	Drainage	D 10
6	Drainage	D 05
7	Electrolyte, Cu	E 02
8	Gasol / Propane	G 02
9	Oxygen	G 35
10	Compressed Air	G 75
11	Instrument Air oil free	G 77
12	Compressed Air, Process	G 78
13	Process Ventilation Gas, Dry	G 90
14	Process Ventilation Gas	G 91
15	Process Ventilation Gas, Wet	G 93
16	Process Ventilation Gas, Wet	G 95
17	Inhibitor Solution	K 13
18	Polymer Solution	K 53
19	Sulphuric Acid	K 90
20	Sulphuric Acid, Max. 70% (Black Acid)	K 91
21	Granulation Water	M 06
22	Heavy Fuel, Oil	O 05
23	Diesel	O 06
24	Steam	S 02
25	Steam, Process	S 15
26	Process Cooling Water	W 24
27	Cooling Water	W 35
28	Warm Water	W 37
29	All Equipments	-
30	Equipments with Hazardous Fluids	-

Existing Media Code / Legend for Above Ground & Under Ground Piping

Sr. No.	Content	Media Code
1	Drinking Water	WDK
2	De-Mineralized Water	DM
3	Condensate	CL
4	Propane	LPG
5	Oxygen	OX
6	Plant Air	AP
7	Compressed Air / Instrument Air	AI
8	Sulphuric Acid	SA
9	Fuel Oil	FO
10	Light Diesel Oil	LDO
11	Low Pressure Steam	SL
12	High Pressure Steam	SM
13	Cooling Water Supply	WCS
14	Cooling Water Return	WCR
15	Filtered Water	WF
16	Soft water	SW

TECHNICAL SPECIFICATIONS

FOR

PAINTING

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1.0 SCOPE

This document covers the details & specification for Painting of pipelines / ducts which apply for Normal as well as Corrosive Industrial Environment and shall be followed wherever applicable.

- 1.1 This specification defines the requirements for the following
 - 1.1.1 Surface preparation
 - 1.1.2 Selection & application of Paints on external surfaces of piping.
- 1.2 The Contractor shall obtain necessary approval from Owner / Owner's Representative before Painting of pipes and shall specify if there are any pipes which do not require Painting.
- 1.3 The Contractor shall perform the work on the basis on the specifications provided and the Piping layout drawings and Assembly drawings as indicated in the Tender Specification.
- 1.4 The Contractor shall follow the instructions provided by Paint Manufacturer as far as possible and practical.
- 1.5 Pipes, fittings etc. shall be Painted only after completion of Mechanical works, testing of pipelines, flushing and only after the approval of Owner / Owner's Representative.
- 1.6 The following shall be ensured:
 - 1.6.1 Proper Storage to avoid exposure or contamination of Paint.
 - 1.6.2 Surface preparation shall be prior to Painting and made to be inspected by Owner / Owner's Representative if required.
 - 1.6.3 Application of Paint and its Drying time etc. shall be as per the recommendations from the Paint Manufacturer.
- 1.7 Any Painting work including surface preparation on piping shall commence only after the system tests have been completed and the piping system is certified for normal function.
- 1.8 The Contractor shall produce Test Reports / Certificates from Manufacturer regarding the quality of the particular batch of Paint supplied for Quality Assurance.

2.0 CODES & STANDARDS

2.1 Surface preparation standards:

The following standards shall be followed for surface preparation wherever applicable:

- 2.1.1 IS-5 – Colors for ready mixed Paints and enamels
- 2.1.2 IS-161 – Heat resistant Paints.
- 2.1.3 IS-2074 – Specification for ready mixed Paint, red oxide zinc chrome priming.
- 2.1.4 IS-2379 – Color code for identification of pipelines
- 2.1.5 IS-2932 – Specification for enamel, synthetic, exterior
(a) Undercoating, (b) Finishing.
- 2.1.6 SSPC codes shall be used where applicable and required.
- 2.1.7 Swedish Standard SIS-05-5900-1967 (Surface preparation standards for Painting Steel Surfaces) shall be applicable for Surface Cleaning Operations.

3.0 TOOLS / EQUIPMENTS FOR SURFACE PREPARATION & PAINTING ACTIVITIES

- All equipments (tools , brushes , rollers, spray guns , blast material , hand power tools for cleaning , scaffolding material , shot/sand blasting equipments & air Compressors etc.) required for carrying out Painting shall be supplied by the Contractor on their own cost .
- Mechanical mixing shall be used for Paint mixing operations in case of two pack systems. The Owner / Owner's representative may allow the hand mixing of small quantities at his discretion.

4.0 SURFACE PREPARATION

4.1 In order to achieve optimum possible surface finish before Painting one must adopt the following methods depending on the condition of pipes/material and instructed by the Engineer-in-Charge or Owner / Owner's Representative:

4.1.1 Manual or hand tool cleaning

It shall only be used where due to safety problems other methods are not feasible. It consists of the following:

- Hand de-scaling and/or hammering
- Hand scraping
- Hand wire brushing

Rust, mill scale spatters, old coatings and other foreign matter, shall be removed by hammering, scrapping tools, emery paper cleaning, wire brushing or combination of the above methods.

On completion of cleaning, loose material shall be removed from the surface by clean rags and the surface shall be brushed, swept, dusted and blown off with compressed air/steam to remove all loose matter. Finally the surface may be washed with water and dried for effective cleaning.

4.1.2 Mechanical or power tool cleaning

It shall be done by the following tools

- Mechanical striking tools
- Chipping hammers
- Grinding Wheels
- Rotating Steel wire brushes

After completion of cleaning procedure loose material on the surface shall be removed using a micro fiber cleaning cloth and compressed air shall be used to remove all loose matters. After all above procedure are completed surface shall be washed using water (recommended) or any other solution and dried.

4.1.3 Blast cleaning

Air blast cleaning is preferred due to existing corrosion on the pipeline but necessary arrangements have to be made by the Contractor to ensure no environmental hazard at the location

- The surfaces shall be blast cleaned during using one of the abrasives:
 - ◆ Sand
 - ◆ Chilled cast iron
 - ◆ Malleable iron and steel
 - ◆ If any other abrasive is being used, the approval of the Owner shall be taken.
- Blast Cleaning shall be carried out at appropriate distance and angle depending on nozzle size maintaining constant velocity and pressure. Chilled cast iron, malleable iron and steel shall be in form of shot or grit whose size will not be greater than 0.055" in case of steel and malleable iron and 0.04" in case of chilled iron.
- Compressed air shall be free from moisture and oil.
- The blasting nozzles should be venturi type with tungsten carbide or boron carbide as the materials for liners.
- Nozzle orifices may vary from 3/16" to 3/4 ".

- On completion of blasting operation, blast surface shall be cleaned and it is to be ensured that the surface is free from any scale or rust and must show a grey white metallic luster.
 - Primer or first coat of Paint shall be applied within 4 hours of surface preparation.
 - Blast cleaning shall never be carried out outdoors in bad weather without adequate protection or when the ambient temperature is less than the dew point. Surface profile shall be uniform to provide good key to the Paint adhesion (i.e. 30 to 50 microns). If possible vacuum collector shall be installed for collecting the abrasives and recycling.
- 4.2 Mill scale, rust, rust scale and foreign matter shall be removed fully to ensure that a clean and dry surface is obtained. The minimum acceptable standards are
- 4.2.1 Manual or hand cleaning shall be ST. 2 as per Swedish Standard SIS-05-5900-1967
 - 4.2.2 Mechanical or power tool cleaning it shall be ST. 3 as per Swedish Standard SIS-05-5900-1967
 - 4.2.3 For Blasting it shall be SA 2-1/2 as per Swedish Standard SIS-05-5900-1967
 - 4.2.4 Highly corrosive condition – Blast cleaning shall be SA 3 as per Swedish Standard SIS-05-5900-1967
- 4.3 Remove all other contaminants, oils grease etc. by use of an aromatic solvent prior to surface cleaning.
- 4.4 After the completion of surface preparation the first coat of primer shall be immediately applied or may be delayed due to unfavorable conditions at the sole discretion of OWNER or engineer in charge.
- Note: Blasting and Painting shall be avoided during unfavorable conditions*

5.0 COATING PROCEDURE AND APPLICATION

- 5.1 Coating operation shall not be carried out in rain, wind or on moist surface or in environment where injurious airborne elements exists, when the steel surface temperature is less than 5°F above dew point, when the relative humidity is greater than 85% or when the temperature is below 40°F. De-humidifier equipment shall be used to control the RH and Dew point for proper curing of internal coating of storage tanks.

- 5.2 Coating operation shall start as soon as practicable but in no case later than 4 Hrs. after one complete application of Primer.
- 5.3 Each coat material shall be applied as a continuous film of uniform thickness to the maximum extent possible. Any spots or areas missed in application shall be recoated and permitted to dry before next coat is applied. Desired wet film thickness shall be maintained while applying Paint.
- 5.4 Each coat shall be in proper state of cure or dryness before the application of succeeding coat. Material shall be considered dry for recoating when an additional coat can be applied without the development of any detrimental film irregularities, such as lifting or loss of adhesion of the under coat. Manufacturer instruction shall be followed for inter coat interval.
- 5.5 When the successive coat of the same colour have been specified, alternate coat shall be tinted, when practical, sufficiently to produce enough contrast to indicate complete coverage of the surface. The tinting material shall be compatible with the material and not detrimental to its service life.
- 5.6 Air spray application shall be in accordance with the following:
- The equipment used shall be suitable for the intended purpose, shall be capable of properly atomizing the Paint to be applied and shall be equipped with suitable pressure regulators and gauges. The air caps, nozzles, and needles shall be those recommended by the Manufacturer of the equipment for the material being sprayed. The equipment shall be kept in satisfactory condition to permit proper Paint application.
 - Traps or separators shall be provided to remove oil and condensed water from the air. These traps or separators must be of adequate size and must be drained periodically during operations. The air from the spray gun impinging against the surface shall show no condensed water or oil.
 - Ingredients shall be kept properly mixed in the spray pots or containers during application by continuous mechanical agitation.
 - The pressure on the material in the pot and of the air at the gun shall be adjusted for optimum spraying effectiveness. The pressure on the material in the pot shall be adjusted when necessary for changes in elevation of the gun above the pot. The atomizing air pressure at the gun shall be high enough to properly atomize the Paint but not so high as to cause excessive fogging of Paint, excessive evaporation of solvent, or loss by over spray.
 - Spray equipment shall be kept sufficiently clean so that dirt, dried Paint, and other foreign materials are not deposited on the Paint film. Any solvents left in the equipment shall be completely removed before applying Paint to the surface being painted.
 - Paint shall be applied in a uniform layer, with overlapping at the edge of the spray pattern. The spray patterns shall be adjusted so that the Paint is deposited uniformly. During application, the gun shall be held perpendicular to the surface and at a distance, which will ensure that a wet layer of Paint is deposited on the surface. The trigger of the gun should be released at the end of each stroke.
 - All runs and sags shall be brushed out immediately or the Paint shall be removed and the surface repainted.
 - Areas inaccessible to the spray gun shall be painted by brush; if not accessible by brush, daubers shall be used.

- All nameplates, Manufacturer's identification tags, finished flange faces, control valve items and similar items shall be masked to prohibit coating deposition. If these surfaces are coated, the component shall be cleaned and restored to its original condition.
- Edges of structural shapes and irregular coated surfaces shall be coated first and an extra pass made later.
- If spray gun shows choking, immediately de-choking procedure shall be followed.

5.7 Brush application of Paint shall be in accordance with the following:

- Brushes shall be of a style and quality that will enable proper application of Paint.
- Rounds or oval brushes are most suitable for rivets, bolts, irregular surface, and rough or pitted steel. Wide flat brushes are suitable for large flat areas, but they shall not have width over five inches.
- Paint shall be applied into all corners.
- Any runs or sags shall be brushed out.
- There shall be a minimum of brush marks left in the applied Paint.
- Surfaces not accessible to brushes shall be painted by spray and daubers.

5.8 Manual application by sling (where 6 O' clock position of pipe is not approachable)

- A canvas strip about 450 mm wide and 1.5m long is held under the pipe by two men from the top in a sling-like position
- It is then moved up and down and slowly moved forward while fresh coating is poured on top of the pipe.
- The men manipulate the sling so that an even coating is obtained all round the bottom. This work shall be done very carefully and by experienced personnel. There shall not be any formation of "Whiskers" and holes in the coating. The coating film shall be inspected at the bottom by mirror.

5.9 For each coat the Painter should know the WFT corresponding to the specified DFT and standardize the Paint application technique to achieve the desired WFT. This has to be ensured in the qualification trial.

6.0 IDENTIFICATION SIGNS

The followings shall be marked:

- Flow direction –indicated by black or white arrow according to contrast at battery limit , intersection points & and change in direction points, midways of piping , near valves, at start and terminating points and at intervals of 50 M for yard piping.
- Product (Service/Fluid/Media) names & Media codes shall be marked at pump inlet, outlet, and midways of piping & battery limit in a suitable size.
- Color Bands shall be used to indicate the battery limits for piping system and also to indicate any hazardous (Black and Golden Band) service in the pipe line. The color bands shall be used in the same order as the direction of flow.

7.0 DRYING OF COATED SURFACES

- 7.1 Next coat shall be applied only after the preceding coat dries off completely. The material shall be considered dry for re-coating when another coat can be applied without the development of any film irregularities such as lifting or loss of adhesion of undercoats. Drying time of the applied coat should not exceed maximum specified for it as a first coat; if it exceeds the Paint material has possibly deteriorated or mixing is faulty.
- 7.2 No Paint shall be force dried under conditions, which will cause cracking, wrinkling, blistering formation of pores, or detrimentally affect the conditions of the Paint.
- 7.3 No external drier shall be added to Paint on the job unless specifically called for in the Manufacturer's specification for the Paint.
- 7.4 Paint shall be protected from rain, condensation, contamination, snow and freezing until dry to the maximum extent possible.

8.0 EXCLUSION(S):

- 8.1 The following are not to be Painted and must be protected from Painting:
 - Porcelain
 - Gauge Glasses
 - Meter Faces
 - Valve Stem and Threads
 - Working Surfaces of Equipment
 - Name Plates
 - Non-Ferrous, Plastic and Stainless Steel (But respective Color Coding have to be provided)
 - Insulation or Fireproofing
 - Galvanized Steel
 - Concrete, Masonry etc.
 - Factory Finished Control Panels
 - Factory Finished Instrument Cases and Meters
 - Factory Finished Equipment, Including Filters, Control Valves etc.

- 8.2 Masking or compounds used to protect non-Painted surface must be removed prior to completion of the job.

9.0 INSPECTION & TESTING

- All Painting materials including primers, thinners & Paints shall be bought directly from the Manufacturer and the Manufacturer's Test certificate shall be produced by the Contractor before starting work.
- Engineer in-charge/Owner shall test the Dry film thickness after each coat to ensure the minimum requirements.
- The Contractor shall provide equipment for inspection of Dry film thickness

10.0 DOCUMENTATION

The Painting Agency shall provide / maintain the following:

- A written quality plan with procedure for qualification trials and for the actual work.
- Daily progress report with details of weather conditions, particular of applications, no of coats and type of materials applied, anomalies, progress of work versus program.
- Results of measurement of temperature, relative humidity, surface profile, film thickness, adhesion tests with signature of appropriate authority.
- Particulars of surface preparation and Paint application during trials and during the work.
- Details of non-compliance, rejects and repairs.
- Type of testing equipments and calibration.
- Code and batch numbers of Paint materials used.

11.0 COLOR CODING SCHEME FOR PIPES

The following Color Code shall be followed for different pipes as mentioned below:

Sr. No.	Content	Ground Color	First Color Band	Second Color Band
1	Drinking Water	Sea Green	French Blue	Signal Red
2	De-Mineralized Water	Sea Green	Light Orange	-
3	Condensate	Sea Green	Light Brown	-
4	Condensate, Oxide	Sea Green	Light Brown	-
5	Drainage	Black	-	-
6	Drainage	Black	-	-
7	Electrolyte, Cu	Black	-	-
8	Gasol / Propane	Canary Yellow	Signal Red	Light Brown
9	Oxygen	Canary Yellow	White	-
10	Compressed Air	Sky Blue	French Blue	-
11	Instrument Air oil free	Sky Blue	French Blue	-
12	Compressed Air,Process	Sky Blue	Silver Grey	-

13	Process Ventilation Gas, Dry	Canary Yellow	White	Dark Violet
14	Process Ventilation Gas	Canary Yellow	White	Dark Violet
15	Process Ventilation Gas, Wet	Canary Yellow	White	Dark Violet
16	Process Ventilation Gas, Wet	Canary Yellow	White	Dark Violet
17	Inhibitor Solution	Black	-	-
18	Polymer Solution	Black	-	-
19	Sulphuric Acid	Dark Violet	Brilliant Green	Light Orange
20	Sulphuric Acid, Max. 70% (Black Acid)	Dark Violet	Brilliant Green	Light Orange
21	Granulation Water	Sea Green	Light Grey	-
22	Heavy Fuel, Oil	Light Brown	-	-
23	Diesel	Light Brown	-	-
24	Steam	Aluminium Paint as per IS 2339	Canary Yellow	-
25	Steam, Process	Aluminium Paint as per IS 2339	Silver Grey	
26	Process Cooling Water	Sea Green	French Blue	-
27	Cooling Water	Sea Green	French Blue	-
28	Warm Water	Sea Green	Canary Yellow	-
29	All Equipments	Light Grey	-	-
30	Equipments with Hazardous Fluids	Canary Yellow	-	-

12.0 PAINT SYSTEMS FOR CORROSIVE INDUSTRIAL ENVIRONMENT

For Corrosive Environment following Paint systems will be applied:

Description	Temp. Range (in °C)	Surface preparation	Primer	Final Coat	Total DFT MICRON (Min)	Remarks
Corrosive Industrial Environment						
External surface of pipes, Tanks, Vessels etc.	-10 to 20	SSPC-SP-3	Two coats High build Chlorinated Rubber Zinc Phosphate Primer – 50 micron/coat (Min)	Two coats of Chlorinated Rubber Paint – 30 micron/coat (Min)	160	Primer & Paint application at ambient temp.
	21 to 80	SSPC-SP-10	Two coats Epoxy Zinc Chromate Primer-35 micron/coat (Min)	Two coats of Epoxy High Build Coating – 100 micron/coat (Min)	270	Paint application at ambient temperature
	81 to 400	SSPC-SP-3	Covered in Finish Coat	Three coats of Heat Resistant Silicon Paint suitable upto 400 °C – 20 Micron / Coat (Min)	60	Paint application at ambient temperature and curing at 250 °C for 4 Hrs.

13.0 PAINT SPECIFICATIONS FOR NORMAL INDUSTRIAL ENVIRONMENTS:

Following Paint systems shall be applied on in Normal Industrial Environments as per requirements:

Description	Temp. Range (in °C)	Surface preparation	Primer	Final Coat	Total DFT MICRON (Min)	Remarks
Normal Industrial Environment						
External surface of pipes, Tanks, Vessels etc.	-10 to 20	SSPC-SP-3	One coat of High Build Chlorinated Rubber Zinc Phosphate Primer – 50 micron/coat (Min)	One coat of High Build Chlorinated Rubber Micaceous Iron Oxide Paint – 100 Microns / coat (Min) Two coats of Chlorinated Rubber Paint – 30 Microns / Coat (Min)	210	Primer & Paint application at ambient temp.
	21 to 60	SSPC-SP-06	Two coats of Red Oxide Zinc Chromate Primer-25 micron/coat (Min)	Two coats of Synthetic Enamel – 20 micron / coat (Min)	90	Paint application at ambient temperature
	61 to 80	SSPC-SP-06	Two coats of High Build Zinc Phosphate Primer – 50 micron / coat (Min)	Two coats of Synthetic Rubber Based Aluminium Paint suitable upto 150 °C – 25 micron / coat (Min)	150	Paint application at ambient temperature
	81 to 250	SSPC-SP-06	Covered in finish coat	Three coats of Heat resistant Aluminium Paint suitable upto 250 °C – 20 Micron / coat (Min)	60	Paint application at ambient temp, curing at elevated temp during start up
	251 to 400	SSPC-SP-10	Covered in Finish Coat	Three coats of Heat Resistant Silicon Paint suitable upto 400 °C – 20 Micron / Coat (Min)	60	Paint application at ambient temperature and curing at elevated Temperatures.

14.0 PRIMER & FINISH COAT MATERIAL SPECIFICATIONS:

PRIMER(S):	
A. Red Oxide Zinc Chromate Primer	
Type	Single Pack
Composition	Modified Phenolic alkyd Medium pigmented with Red Oxide and Zinc Chromate
Volume solids	35%
B. High Build Chlorinated Rubber Zinc Phosphate Primer	
Type	Single Pack
Composition	Chlorinated Rubber Medium Plasticised with un-saponifiable plasticizer pigmented with Zinc Phosphate
Volume solids	40%
C. High Build Phosphate Primer	
Type	Single Pack
Composition	Synthetic Medium, pigmented with Zinc Phosphate
Volume solids	40%
D. Epoxy Zinc Chromate Primer	
Type	Two Pack
Composition	Polyamide cured Epoxy resin medium pigmented with Zinc Chromate
Volume solids	40%
FINISH COAT(S):	
A. Synthetic Enamel	
Type	Single Pack
Composition	Alkyd Medium pigmented with superior quality water and weather resistant pigments

Volume solids	40%
B. Chlorinated Rubber Paint	
Type	Single Pack
Composition	Plasticised Chlorinated Rubber medium with chemical and weather resistant pigments
Volume solids	30%
C. High Build Chlorinated Rubber Micaceous Iron Oxide Paint	
Type	Single Pack
Composition	Chlorinated Rubber based High build pigmented with Micaceous Iron Oxide
Volume solids	40%
D. Epoxy High Building Coating	
Type	Two Pack
Composition	Polyamide cured epoxy resin medium suitably pigmented
Volume solids	60%
E. Heat Resistant Aluminium Paint suitable upto 250 °C	
Type	Dual Container
Composition	Heat resistant special varnished medium combined with Aluminium flakes.
Volume solids	20%
F. Heat Resistant Silicon Paint suitable upto 400 °C	
Type	Single Pack
Composition	Silicon resin based with Aluminium flakes
Volume solids	20%
G. Synthetic Rubber based Aluminium Paint suitable upto 150 °C	
Type	Single Pack
Composition	Synthetic Rubber medium combined with leafing Aluminium

PIPING MATERIAL SPECIFICATION

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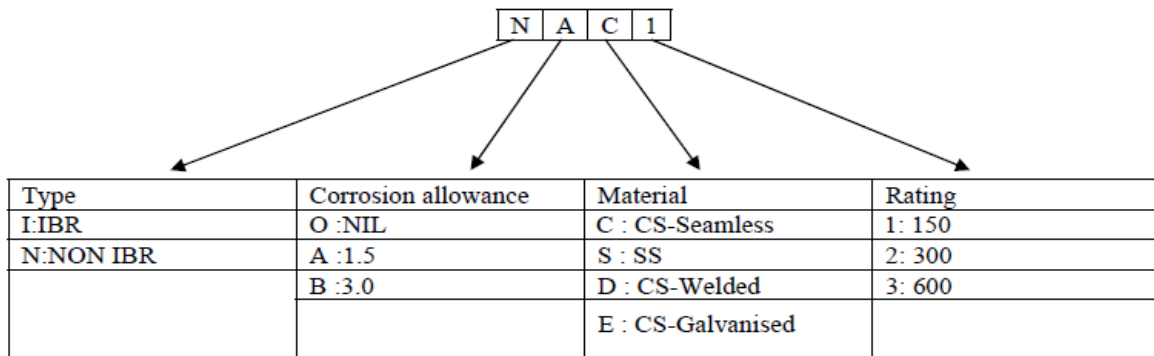
Section 1.0

Pipe Specification by Package Vendor & Equivalent Pipe Specification by MCS (Table 1.2)

Pipe Specification By Package Vendor		Equivalent Pipe Specification by MCS
A1A	:	NAC1
A2A	:	IAC1
A3A	:	NAD1
J2A	:	NOE1
A6K	:	NAS1
A8A	:	NBC1

Table 1.1

Class Identification (Table 1.2)



Pipe Identification (Table 1.3)

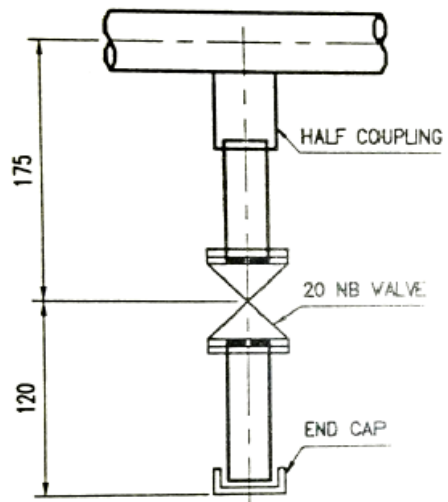
Pipe Specification	Service
NAC1	LOW PRESSURE STEAM, DE MINERALISED WATER, CONDENSATE, FUEL OIL , LIGHT DIESEL OIL, PROPANE,FIRE WATER & WATER SERVICES BEYOND NAD1, NaOH, BRINE
IAC1	HIGH PRESSURE STEAM (IBR), FEED WATER (IBR) , CONDENSATE (IBR)
NAD1	FILTERED WATER, INSTRUMENT AIR, PLANT AIR, COOLING WATER SUPPLY, COOLING WATER RETURN,SOFT WATER, BOILER FEED SUCTION(NON IBR), DILUTED NaOH, AIR
NOE1	DRINKING WATER
NAS1	OXYGEN & OXYGEN + AIR
NBC1	SULPHURIC ACID-(80-98.5% CONC.), SUPHURIC ACID VAPOUR

Section 2.0

TYPICAL VENT & DRAIN DETAILS

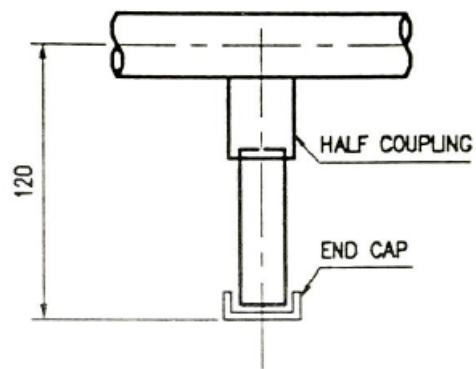
DRAIN CONNECTION DETAILS FOR WATER & OIL LINES

(Header Size DN 50 & Above)



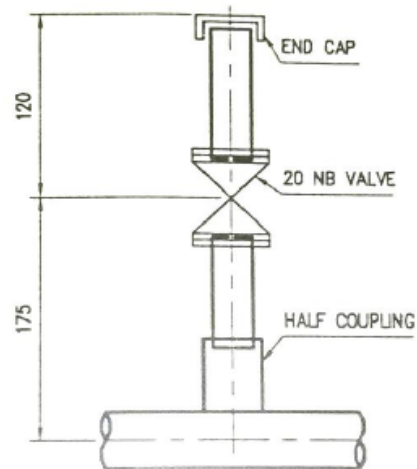
DRAIN CONNECTION DETAILS FOR GAS LINES

(Header Size DN 50 & Above)



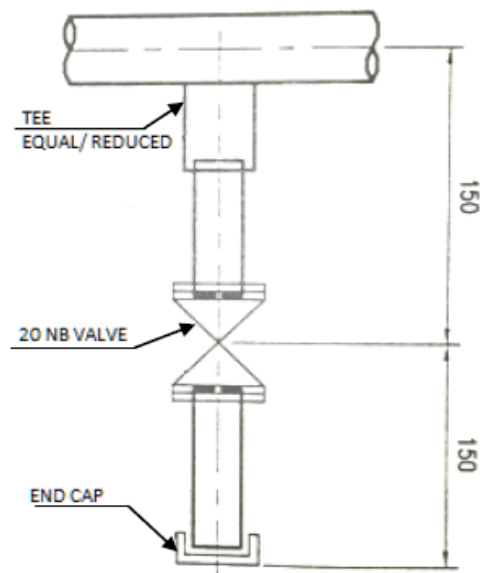
VENT CONNECTION DETAILS FOR WATER, GAS, OIL & STEAM LINES

(Header Size DN 50 & Above)



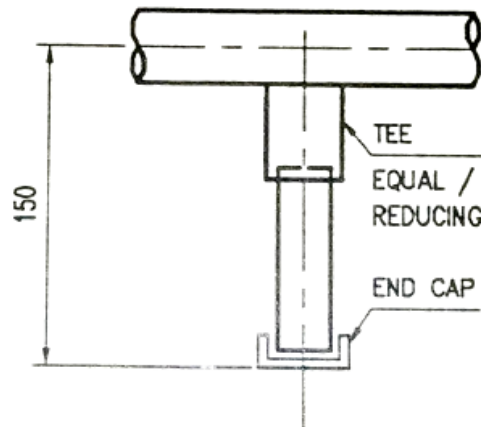
DRAIN CONNECTION DETAILS FOR WATER & OIL LINES

(Header Size DN 40 & Above)



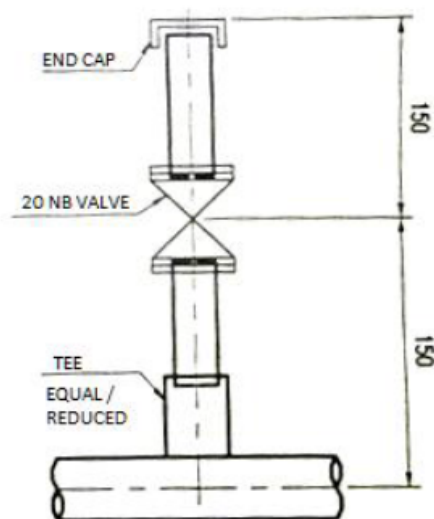
DRAIN CONNECTION DETAILS FOR GAS LINES

(Header Size DN 40 and Below)



VENT CONNECTION DETAILS

(Header Size DN 40 & Below)



Notes: * All Dimensions are in mm. unless otherwise stated

*Vents and Drains shall be provided with valves as per Piping Material Specification.

	METALTEC	PIPING MATERIAL SPECIFICATION					
	CONSULTANCY & SERVICES	PIPE CLASS :NAC1					
	202A,Earth-The Landmark						
	Sun Pharma Road,Atladra,Vadodara,India						
DOC. NO : MCS / PMS / 01-06			0	18.06.2015	SSH	SK	-
SHEET : 1 OF 2			REV.	DATE	PPD BY	CHKD BY	REMARKS

PIPE CLASS	:	NAC1																														
FLANGE RATINGS	:	ASA 150 #																														
BASE MATERIAL	:	CARBON STEEL																														
CORROSION ALLOWANCE	:	1.5 MM																														
TEMPERATURE	:	200°C Max.																														
PRESSURE	:	14 kg/cm ² g (Max.)																														
SERVICES	:	LOW PRESSURE STEAM, DE MINERALISED WATER*, CONDENSATE, FUEL OIL , LIGHT DIESEL OIL, PROPANE , FIRE WATER & WATER SERVICES BEYOND NAD1,NaOH, BRINE																														
NOTES	:	FLANGES ARE AS PER ANSI B 16.5, ASA 150 #, RAISED FACE. NUT BOLTS ARE STANDARD METRIC THREADED HOT DIP GALVANIZED AS PER IS 1367 Cl. 4 6/4 ALL DRAINS AND VENTS ARE AS PER SECTION 2.0 MITER JOINTS SHALL BE AS PER ASME B31.3 , MOC FOR MITER IS ASTM A672 GRB60 CL12 * RUBBER LINING (WHERE APPLICABLE) SHALL BE OF HARD EBONITE RUBBER (4MM THICK) 'M' DENOTES MATCH TO PIPE THICKNESS																														
SPECIAL NOTES	:	<table border="1"> <thead> <tr> <th>ITEMS</th> <th>SIZE</th> <th>DESCRIPTION</th> </tr> </thead> <tbody> <tr> <td>MAINTENANCE JOINTS</td> <td>ALL</td> <td>FLANGED, TO BE KEPT MINIMUM</td> </tr> <tr> <td>PIPE JOINTS</td> <td>40 MM & BELOW</td> <td>SOCKET WELD COUPLING</td> </tr> <tr> <td></td> <td>50 MM & ABOVE</td> <td>BUTT WELDED</td> </tr> <tr> <td>DRAINS</td> <td>ON LINE <= 40 MM</td> <td>Refer Section 2.0</td> </tr> <tr> <td></td> <td>ON LINE >= 50 MM</td> <td>Refer Section 2.0</td> </tr> <tr> <td>VENTS</td> <td>ON LINE <= 40 MM</td> <td>Refer Section 2.0</td> </tr> <tr> <td></td> <td>ON LINE >= 50 MM</td> <td>Refer Section 2.0</td> </tr> <tr> <td>TEMP. CONNECTION</td> <td>40 MM & BELOW</td> <td>FLANGED</td> </tr> <tr> <td>PRESS. CONNECTION</td> <td>20 MM</td> <td>SOCKET WELD WITH GATE VALVE</td> </tr> </tbody> </table>	ITEMS	SIZE	DESCRIPTION	MAINTENANCE JOINTS	ALL	FLANGED, TO BE KEPT MINIMUM	PIPE JOINTS	40 MM & BELOW	SOCKET WELD COUPLING		50 MM & ABOVE	BUTT WELDED	DRAINS	ON LINE <= 40 MM	Refer Section 2.0		ON LINE >= 50 MM	Refer Section 2.0	VENTS	ON LINE <= 40 MM	Refer Section 2.0		ON LINE >= 50 MM	Refer Section 2.0	TEMP. CONNECTION	40 MM & BELOW	FLANGED	PRESS. CONNECTION	20 MM	SOCKET WELD WITH GATE VALVE
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TEMP. CONNECTION	40 MM & BELOW	FLANGED																														
PRESS. CONNECTION	20 MM	SOCKET WELD WITH GATE VALVE																														

Elbow	ASME B16.9, B16.11
Fittings (Tees,Reducers, etc.)	ASME B16.9, B16.11
Flanges / Blind Flange	ANSI B 16.5
Bolting	ASME B18.2.1(BOLTS),ANSI B18.2.2 (NUTS)
Gasket	ASME B 16.21
Strainer	ALL STRAINER SHALL BE HYDROSTATICALLY TESTED AT TWICE THE DESIGN PRESSURE

	METALTEC		PIPING MATERIAL SPECIFICATION								
	CONSULTANCY & SERVICES										
	202A, Earth-The Landmark										
	Sun Pharma Road, Atladra, Vadodra, India										
	DOC. NO : MCS/PMS/01-06										
SHEET : 2 OF 2						0	18.06.2015	SSH	SK	-	
						REV.	DATE	PPD BY	CHKD BY	REMARKS	
Item	Description	Diameter Range (NB)		SCH/ Thk.(mm) /Rating	Dimensional Standard	Material	Remarks				
		From	To								
PIPES											
	SEAMLESS	15	40	Sch. 80	B-36.10	API 5L GR. B	PLAIN END				
	SEAMLESS	50	150	Sch. 40	B-36.10	API 5L GR. B	BEVELLED END				
	SEAMLESS	200	300	Sch. 20	B-36.10	API 5L GR. B	BEVELLED END				
FLANGE											
	SOCKET WELD FLANGE	15	40	150	ANSI B 16.5	ASTM A 105	RAISED FACE				
	WELD NECK FLANGE	50	300	150	ANSI B 16.5	ASTM A 105	RAISED FACE				
	BLIND	15	300	150	ANSI B 16.5	ASTM A 105	RAISED FACE				
FITTINGS											
ELBOW											
	ELBOW 45° & 90°	15	40	3000	ASME B 16.11	ASTM A 105	SOCKET WELD				
	ELBOW 45° & 90°	50	300	M	ASME B 16.9	ASTM A 234 GR. WPB	BEVELLED END				
TEE											
	EQUAL & REDUCING	15	40	3000	ASME B 16.11	ASTM A 105	SOCKET WELD				
	EQUAL & REDUCING	50	300	M	ASME B 16.9	ASTM A 234 GR. WPB	BEVELLED END				
REDUCER											
	SWAGE REDUCER	15	40	3000	ASME B 16.11	ASTM A 105	PLAIN END				
	CONCENTRIC	15	300	M	ASME B 16.9	ASTM A 234 GR. WPB	BEVELLED END				
	ECCENTRIC	15	300	M	ASME B 16.9	ASTM A 234 GR. WPB	BEVELLED END				
END CAP											
	SCREWED NPT	15	40	3000	ASME B 16.11	ASTM A 105					
	UNION	15	40	M	ASME B 16.11	ASTM-A107 Gr 2					
	CAP	50	300	M	ASME B 16.9	ASTM A 234 GR. WPB, SCH40	BEVELLED END				
COUPLING - FULL & HALF											
	COUPLING	15	40	3000	ASME B 16.11	ASTM A 105	SOCKET WELD				
VALVE											
	GATE VALVE	15	40	800	API-602	BODY-ASTM A 105, TRIM-13% Cr STEEL	SOCKET WELD				
	GATE VALVE	50	300	150	API-600	BODY-ASTM A 216 GR. WCB., TRIM- 13% Cr STEEL	FLANGED END , B-16.5, RAISED FACE				
	GLOBE VALVE	15	40	800	BS-5352	BODY-ASTM A 105, TRIM-13% Cr STEEL	SOCKET WELD				
	GLOBE VALVE	50	300	150	BS-1873	BODY-ASTM A 216 GR. WCB., TRIM- 13% Cr STEEL	FLANGED END , B-16.5, RAISED FACE				
	CHECK VALVE	15	40	800	BS-5352	BODY-ASTM A 105, TRIM -13% CR	SOCKET WELD				
	CHECK VALVE	50	300	150	BS-1868	BODY-ASTM A 216 GR. WCB., TRIM- 13% Cr STEEL	FLANGED END , RAISED FACE				
	BALL VALVE	15	300	150	BS-5351	BODY-ASTM A 216 GR.WCB , RP1TF SEAT	RAISED FACE				
	PLUG VALVE	15	300	150	BS-5353	BODY- A 216 GR.WCB, TRIM-PLUG-HARDENED	FLANGED END, ASME B-16.5				
	BUTTERFLY VALVE	80	300	150	BS-5155	BODY- A 216 GR.WCB., TRIM-13% Cr STEEL	WAFER TYPE				
NUTS & BOLTS											
	BOLTS/NUTS	-	-	-	ASME B 18.2/B 18.22	BOLT-A193 GR. B7, NUT-A194 GR. 2H					
GASKETS											
	GASKET	15	300	2 MM	ASME B 16.21	IS-2712-GR.O1					
TRAP/STRAINER											
	STEAM TRAP	15	40	150	-	BODY-ASTM A 105, TRAP-13% Cr INTERNAL-SS 304	FLANGED, THERMODYNAMIC TRAP, 150 CLASS, RAISED FACE				
	TEMPORARY STRADNER	15	300	150	-	BODY - ASTM A516GR70, INT-SS304	FLAT FACED				
	PERMANENT STRADNER	15	40	800	-	BODY-ASTM A 105, INTERNAL-SS 304	SOCKET WELD, Y-TYPE, 800				
	PERMANENT STRADNER	50	300	150	-	BODY-ASTM A 216 GR.WCB., INTERNAL - SS 304	BEVELLED END, T-TYPE				

	METALTEC	PIPING MATERIAL SPECIFICATION					
	CONSULTANCY & SERVICES	PIPE CLASS : IAC1					
	202A,Earth-The Landmark						
	Sun Pharma Road,Atladra,Vadodara,India						
DOC. NO : MCS/PMS/02-06			0	18.06.2015	SSH	SK	-
SHEET : 1 OF 2			REV.	DATE	PPD BY	CHKD BY	REMARKS

PIPE CLASS	:	IAC1																														
FLANGE RATINGS	:	ASA 150 #																														
BASE MATERIAL	:	CARBON STEEL																														
CORROSION ALLOWANCE	:	1.5 MM																														
TEMPERATURE	:	200°C Max.																														
PRESSURE	:	14 kg/cm ² g (Max.)																														
SERVICES	:	HIGH PRESSURE STEAM (IBR) , FEED WATER (IBR) , CONDENSATE (IBR)																														
NOTES	:	FLANGES ARE AS PER ANSI B 16.5, ASA 150 #, RAISED FACE. NUT BOLTS ARE STANDARD METRIC THREADED HOT DIP GALVANIZED AS PER IS 1367 CL. 4 6/4 ALL DRAINS AND VENTS ARE AS PER SECTION 2.0 'M' DENOTES MATCH TO PIPE THICKNESS																														
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Elbow	ASME B 16.9, B 16.11
Fittings (Tees,Reducers, etc.)	ASME B 16.9, B 16.11
Flanges / Blind Flange	ANSI B 16.5
Bolting	ASME B 18.2.1,ANSI B18.2.2
Gasket	ASME B 16.21

	METALTEC CONSULTANCY & SERVICES 202A,Earth-The Landmark Sun Pharma Road,Atladra,Vadodara,India	PIPING MATERIAL SPECIFICATION								
	DOC. NO : MCS/PMS/02-06 SHEET : 2 OF 2			PIPE CLASS :IAC1						
					0	18.06.2015	SSH	SK		
					REV.	DATE	PPD BY	CHKD BY	REMARKS	

Item	Description	Diameter Range		SCH/ Thk. (mm)/Rating	Dimensional Standard	Material	Remarks
		From	To				
PIPES							
	SEAMLESS	15	40	Sch. 80	ASME B 36.10	ASTM A 106 GR.B	FLAT END
	SEAMLESS	50	150	Sch. 40	ASME B 36.10	ASTM A 106 GR.B	BEVELLED END
	SEAMLESS	200	300	Sch. 20	ASME B 36.10	ASTM A 106 GR.B	BEVELLED END
FLANGE							
	SOCKET WELD FLANGE	15	40	150	ANSI B 16.5	ASTM A 105	RAISED FACE
	WELD NECK FLANGE	50	300	150	ANSI B 16.5	ASTM A 105	RAISED FACE
	BLIND	15	300	150	ANSI B 16.5	ASTM A 105	RAISED FACE
FITTINGS							
ELBOW							
	ELBOW 45° & 90°	15	40	3000	ASME B 16.11	ASTM A 105	SOCKET WELD
	ELBOW 45° & 90°	50	300	M	ASME B 16.9	ASTM A 234 GR. WPB	BEVELLED END
TEE							
	EQUAL & REDUCING	15	40	3000	ASME B 16.11	ASTM A 105	SOCKET WELD
	EQUAL & REDUCING	50	150	M	ASME B 16.9	ASTM A 234 GR. WPB	BEVELLED END
	EQUAL & REDUCING	200	300	M	ASME B 16.9	ASTM A 234 GR. WPB	BEVELLED END
REDUCER							
	CONCENTRIC & ECCENTRIC	15	150	3000	ASME B 16.9	ASTM A 234 GR. WPB	BEVELLED END
	CONCENTRIC & ECCENTRIC	200	300	M	ASME B 16.9	ASTM A 234 GR. WPB	BEVELLED END
VALVE							
	GATE VALVE	15	40	800	API-602	BODY-ASTM A 105, TRIM-STELLITEO, STEM-13% Cr STEEL	SOCKET WELD
	GATE VALVE	50	300	150	API-600	BODY-ASTM A 216 GR.WCB., TRIM- 13% Cr STEEL	FLANGED END , B-16.5,RAISED FACE
	GLOBE VALVE	15	40	800	BS-5352	BODY-ASTM A 105, TRIM-STELLITEO, STEM-13% Cr STEEL	SOCKET WELD
	GLOBE VALVE	50	300	150	BS-1873	BODY-ASTM A 216 GR.WCB., TRIM- 13% Cr STEEL	FLANGED END , B-16.5,RAISED FACE
	CHECK VALVE	15	40	800	BS-5352	BODY-ASTM A 105, 13% Cr TRIM	SOCKET WELD
	CHECK VALVE	50	300	150	BS-1868	BODY-ASTM A 216 GR.WCB., TRIM- 13% Cr STEEL	FLANGED END , RAISED FACE
	BALL VALVE	15	300	150	BS-5351	BODY-ASTM A 216 GR.WCB , PEEK SEAT	RAISED FACE
NUTS & BOLTS							
	BOLTS/NUTS	-	-		ASME B 18.2/B 18.22	BOLT-A193 GR. B7, NUT-A194 GR. 2H	
GASKETS							
	GASKET	15	300	2 MM	ASME B 16.21	IS-2712-GR.W/I	
TRAP/STRAINER							
	STEAM TRAP	15	40	150	-	BODY-ASTM A 105, TRAP-13% Cr, INTERNAL-SS 304	FLANGED, THERMODYNAMICS, 150, RAISED FACE
	TEMPORARY STRAINER	15	300	150	-	BODY - ASTM A516GR70, INT:SS304	FLAT FACED
	PERMANENT STRAINER	15	40	800	-	BODY-ASTM A 105, INTERNAL-SS 304	SOCKET WELD, Y-TYPE,800
	PERMANENT STRAINER	50	300	150	-	BODY-ASTM A 216 GR.WCB., INTERNAL - SS 304	BEVELLED END, T-TYPE

	METALTEC	PIPING MATERIAL SPECIFICATION																																			
	CONSULTANCY & SERVICES	PIPE CLASS :NAD1																																			
	202A,Earth-The Landmark																																				
	Sun Pharma Road,Atladra,Vadodara,India																																				
DOC. NO : MCS/PMS/03-06			0	18.06.2015	SSH	SK	-																														
SHEET : 1 OF 2			REV.	DATE	PPD.BY	CHKD.BY	REMARKS																														
PIPE CLASS : NAD1 FLANGE RATINGS : ASA 150 # BASE MATERIAL : CARBON STEEL CORROSION ALLOWANCE : 1.5 MM TEMPERATURE : 150°C Max. PRESSURE : 10.5 kg/cm² g (Max.) SERVICES : FILTERED WATER, INSTRUMENT AIR, PLANT AIR, COOLING WATER SUPPLY, COOLING WATER RETURN,SOFT WATER, DILUTED NaOH, AIR NOTES : FLANGES ARE AS PER ANSI B 16.5, ASA 150 #, RAISED FACE. : NUT BOLTS ARE STANDARD METRIC THREADED HOT DIP GALVANIZED AS PER IS 1367 Cl. 4 6/4 : ALL DRAINS AND VENTS SHALL BE AS PER SECTION 2.0 : PIPE SHALL BE GALVANISED FOR INSTRUMENT AIR : RUBBER LINING (WHERE APPLICABLE) SHALL BE OF HARD EBONITE RUBBER (4MM THICK) : 'M' DENOTES MATCH TO PIPE THICKNESS SPECIAL NOTES : <table><thead><tr><th>ITEMS</th><th>SIZE</th><th>DESCRIPTION</th></tr></thead><tbody><tr><td>MAINTANANCE JOINTS</td><td>ALL</td><td>FLANGED, TO BE KEPT MINIMUM</td></tr><tr><td>PIPE JOINTS</td><td>40 MM & BELOW</td><td>SOCKET WELD COUPLING</td></tr><tr><td></td><td>50 MM & ABOVE</td><td>BUTT WELDED</td></tr><tr><td>DRAINS</td><td>ON LINE <= 40 MM</td><td>Refer Section 2.0</td></tr><tr><td></td><td>ON LINE >= 50 MM</td><td>Refer Section 2.0</td></tr><tr><td>VENTS</td><td>ON LINE <= 40 MM</td><td>Refer Section 2.0</td></tr><tr><td></td><td>ON LINE >= 50 MM</td><td>Refer Section 2.0</td></tr><tr><td>TEMP. CONNECTION</td><td>40 MM & BELOW</td><td>FLANGED</td></tr><tr><td>PRESS. CONNECTION</td><td>20 MM</td><td>SOCKET WELD WITH VALVE</td></tr></tbody></table>								ITEMS	SIZE	DESCRIPTION	MAINTANANCE JOINTS	ALL	FLANGED, TO BE KEPT MINIMUM	PIPE JOINTS	40 MM & BELOW	SOCKET WELD COUPLING		50 MM & ABOVE	BUTT WELDED	DRAINS	ON LINE <= 40 MM	Refer Section 2.0		ON LINE >= 50 MM	Refer Section 2.0	VENTS	ON LINE <= 40 MM	Refer Section 2.0		ON LINE >= 50 MM	Refer Section 2.0	TEMP. CONNECTION	40 MM & BELOW	FLANGED	PRESS. CONNECTION	20 MM	SOCKET WELD WITH VALVE
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Fittings (Tees,Reducers, etc.)		ASME B 16.9, B 16.11																																			
Flanges / Blind Flange		ANSI B 16.5																																			
Bolting		Bolts: A193GR.B7, Nuts: A194 GR.2H																																			
Gasket		ASME B 16.21																																			

	METALTEC CONSULTANCY & SERVICES 202A,Earth-The Landmark Sun Pharma Road,Adina,Valodara,jadia	PIPING MATERIAL SPECIFICATION								
		PIPE CLASS :NAD1								
						0	18.06.2015	SSH	SK	-
DOC. NO : MCS/PMS/03-06 SHEET : 2 OF 2						REV.	DATE	PPD BY	CHKD BY	REMARKS
Item	Description	Diameter Range		SCH/ Thk.(mm) /Rating	Dimensional Standard	Material	Remarks			
		From	To							
PIPES										
	C.WELDED	15	40	HEAVY	IS-1239	IS-1239 BLACK	PLAIN END			
	C.WELDED	50	150	HEAVY	IS-1239	IS-1239 BLACK	BEVELLED END			
	WELDED	200	350	6	IS-3589	IS-3589 GR.330	BEVELLED END			
	WELDED	400	450	8	IS-3589	IS-3589 GR.330	BEVELLED END			
	WELDED	500	600	10	IS-3589	IS-3589 GR.330	BEVELLED END			
FLANGE										
	SOCKET WELD FLANGE	15	40	150	ANSI B 16.5	ASTM A 105	RAISED FACE			
	SLIP ON FLANGE	50	600	150	ANSI B 16.5	ASTM A 105	RAISED FACE			
	BLIND	15	600	150	ANSI B 16.5	ASTM A 105	RAISED FACE			
FITTINGS										
ELBOW										
	ELBOW 45° & 90°	15	40	3000	ASME B 16.11	ASTM A 105	SOCKET WELD			
	ELBOW 45° & 90°	50	150	M	ASME B 16.9	ASTM A 234 GR. WPB	BEVELLED END			
	ELBOW 45° & 90°	200	600	M	ASME B 16.9	ASTM A 234 GR. WPBW	BEVELLED END			
TEE										
	EQUAL & REDUCING	15	40	3000	ASME B 16.11	ASTM A 105	SOCKET WELD			
	EQUAL & REDUCING	50	150	M	ASME B 16.9	ASTM A 234 GR. WPB	BEVELLED END			
	EQUAL & REDUCING	200	600	M	ASME B 16.9	ASTM A 234 GR. WPBW	BEVELLED END			
REDUCER										
	SWAGE REDUCER	15	40	3000	ASME B 16.11	ASTM A 105	PLAIN END			
	CONCENTRIC	15	600	M	ASME B 16.9	ASTM A 234 GR. WPB	BEVELLED END			
	ECCENTRIC	15	600	M	ASME B 16.9	ASTM A 234 GR. WPB	BEVELLED END			
END CAP										
	CAP	15	40	3000	ASME B 16.11	ASTM A 105	SCREWED , NPT			
	CAP	50	600	M	ASME B 16.9	ASTM A 234 GR. WPB	BEVELLED END			
COUPLING - FULL & HALF										
	COUPLING	15	40	3000	ASME B 16.11	ASTM A 105	SOCKET WELD			
VALVE										
	GATE VALVE	15	40	800	API-602	BODY-ASTM A 105, TRIM-13% Cr STEEL	SOCKET WELD			
	GATE VALVE	50	600	150	API-600	BODY-ASTM A 216 GR.WCB, TRIM-13% Cr STEEL	RAISED FACE, FLANGED END			
	GLOBE VALVE	15	40	800	BS-5352	BODY-ASTM A 105, TRIM-13% Cr STEEL	SOCKET WELD			
	GLOBE VALVE	50	600	150	BS-1873	BODY-ASTM A 216 GR.WCB, TRIM-13% Cr STEEL	RAISED FACE, FLANGED END			
	CHECK VALVE	15	40	800	BS-5352	BODY-ASTM A 105,TRIM -13% Cr STEEL	SOCKET WELD			
	CHECK VALVE	50	600	150	BS-1868	BODY-ASTM A 216 GR.WCB, TRIM-13% Cr STEEL	RAISED FACE, FLANGED END			
	BUTTERFLY VALVE	80	600	150	BS-5155	BODY- A 216 GR.WCB, TRIM-13% Cr STEEL	WAFFER TYPE			
NUTS & BOLTS										
	BOLTS/NUTS	-	-	-	ASME B 18.2/B 18.22	BOLT-A193 GR. B7, NUT-A194 GR. 2H				
GASKETS										
	GASKET	15	300	2 MM	ASME B 16.21	IS-2712-GR.W/3				
TRAP/STRAINER										
	TRAP	15	40	150	-	BODY-ASTM A 105, TRAP-13% Cr, INTERNAL-SS 304	THERMODYNAMIC TRAP			
	TEMPORARY STRAINER	15	600	150	-	BODY - ASTM A516GR70, DNT-SS304	CONE TYPE			
	PERMANENT STRAINER	15	40	800	-	BODY-ASTM A 105, INTERNAL-SS 304	Y TYPE			

	METALTEC	PIPING MATERIAL SPECIFICATION					
	CONSULTANCY & SERVICES	PIPE CLASS :NOE1					
	202A,Earth-The Landmark						
	Sun Pharma Road,Atladra,Vadodra,India						
DOC. NO : MCS/PMS/04-06			0	18.06.2015	SSH	SK	
SHEET : 1 OF 2			REV.	DATE	PPD BY	CHKD BY	REMARKS

PIPE CLASS	:	NOE1																														
FLANGE RATINGS	:	ASA 150 #																														
BASE MATERIAL	:	CARBON STEEL (GALVANISED)																														
CORROSION ALLOWANCE	:	NIL																														
TEMPERATURE	:	65°C Max.																														
PRESSURE	:	10.5 kg/cm ² g (Max.)																														
SERVICES	:	DRINKING WATER																														
NOTES	:	FLANGES ARE AS PER ANSI B 16.5, ASA 150 #, RAISED FACE. NUT BOLTS ARE STANDARD METRIC THREADED HOT DIP GALVANIZED AS PER IS 1367 Cl. 4 6/4 ALL DRAINS AND VENTS ARE AS PER SECTION 2.0 ALL INSTRUMENTS CONNECTIONS ARE AS PER P & ID's AND SHOULD BE FLANGE ENDS. 'M' DENOTES MATCH TO PIPE THICKNESS																														
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TEMP. CONNECTION	40 MM & BELOW	FLANGED																														
PRESS. CONNECTION	20 MM	SOCKET WELD WITH VALVE																														

Elbow	ASME B 16.9, B 16.11
Fittings (Tees,Reducers, etc.)	ASME B 16.9, B 16.11
Flanges / Blind Flange	ANSI B 16.5
Bolting	Bolts: A193GR.B7, Nuts: A194 GR.2H
Gasket	ASME B 16.21

	METALTEC CONSULTANCY & SERVICES 202A,Earth-The Landmark Sun Pharma Road,Atladra,Vadodara,india	PIPING MATERIAL SPECIFICATION								
		PIPE CLASS :NOE1								
	DOC. NO : MCS/PMS/04-06 SHEET : 2 OF 2			0	18.06.2015	SSH	SK	-		
		REV.	DATE	PPD.BY	CHKD.BY	REMARKS				
Item	Description	Diameter Range		SCH/ Thk. (mm) / Rating	Dimensional Standard	Material	Remarks			
		From	To							
PIPES										
	C.WELDED	15	150	HEAVY	IS-1239	IS-1239 (GALV)	SCREWED			
FLANGE										
	SCREWED	15	150	150	ANSI B 16.5	ASTM A 105 (GALV.)	FLAT FACE			
	BLIND	15	150	150	ANSI B 16.5	ASTM A 105 (GALV.)	FLAT FACE			
FITTINGS										
ELBOW										
	ELBOW 45° & 90°	15	40	3000	ASME B 16.11	ASTM A 105 (GALV.)	SCREWED			
	ELBOW 90°	50	150	HEAVY	IS-1239II	IS-1239 (PART-II) GALV.	SCREWED			
TEE										
	EQUAL & REDUCING	15	40	3000	ASME B 16.11	ASTM A 105 (GALV.)	SCREWED			
	EQUAL & REDUCING	50	150	HEAVY	IS-1239II	IS-1239 (PART-II) GALV.	SCREWED			
REDUCER										
	CONCENTRIC & ECCENTRIC	15	40	3000	ASME B-16.11	ASTM A 105 (GALV.)	SCREWED			
	CONCENTRIC & ECCENTRIC	50	150	HEAVY	IS-1239II	IS-1239 (PART-II) GALV.	SCREWED			
COUPLING - FULL & HALF										
	COUPLING	15	150	3000	ASME B 16.11	ASTM A 105 (GALV.)	SCREWED			
VALVE										
	GATE VALVE	15	40	CL 2	IS-778	IS:318GR.2, IS:320AL.1	SCREWED			
	GATE VALVE	50	150	P1.6	IS-780	IS210GRFG200, IS320A.1	FLAT FACED			
	GLOBE VALVE	15	40	CL 2	IS-778	IS:318GR.2, IS:320AL.1	SCREWED			
	CHECK VALVE	15	40	CL 2	IS-778	IS:318GR.2, IS:320AL.1	SCREWED			
	CHECK VALVE	50	150	P1.6	IS-5312	IS210GRFG200, IS320A.1	FLAT FACED			
NUTS & BOLTS										
	BOLTS/NUTS	-	-	-	ASME B 18.2/B 18.22	BOLT-A193 GR. B7, NUT-A194 GR. 2H	-			
GASKETS										
	GASKET	15	150	2 MM	ASME B 16.21	BUTYL RUBBER	FULL FACE			

	METALTEC	PIPING MATERIAL SPECIFICATION																																
	CONSULTANCY & SERVICES	PIPE CLASS :NAS1																																
	202A,Earth-The Landmark																																	
	Sun Pharma Road,Atladra,Vadodara,India																																	
DOC. NO : MCS/PMS/05-06			0	18.06.2015	SSH	SK	-																											
SHEET : 1 OF 2			REV.	DATE	PPD.BY	CHKD.BY	REMARKS																											
PIPE CLASS : NAS1 FLANGE RATINGS : ASA 150 # BASE MATERIAL : STAINLESS STEEL (SS 304 L) CORROSION ALLOWANCE : 1.5 MM TEMPERATURE : 100°C Max. PRESSURE : 13.5 kg/cm² g (Max.) SERVICES : OXYGEN & OXYGEN + AIR NOTES : FLANGES ARE AS PER ANSI B 16.5, ASA 150 #, RAISED FACE. NOTES : NUT BOLTS ARE STANDARD METRIC THREADED HOT DIP GALVANIZED AS PER IS 1367 CI. 4 6/4 NOTES : ALL DRAINS AND VENTS ARE AS PER SECTION 2.0 NOTES : 'M' DENOTES MATCH TO PIPE THICKNESS SPECIAL NOTES : <table><thead><tr><th>ITEMS</th><th>SIZE</th><th>DESCRIPTION</th></tr></thead><tbody><tr><td>MAINTANANCE JOINTS</td><td>ALL</td><td>FLANGED, TO BE KEPT MINIMUM</td></tr><tr><td rowspan="2">PIPE JOINTS</td><td>40 MM & BELOW</td><td>SOCKET WELD COUPLING</td></tr><tr><td>50 MM & ABOVE</td><td>BUTT WELDED</td></tr><tr><td rowspan="2">DRAINS</td><td>ON LINE <= 40 MM</td><td>Refer Section 2.0</td></tr><tr><td>ON LINE >= 50 MM</td><td>Refer Section 2.0</td></tr><tr><td rowspan="2">VENTS</td><td>ON LINE <= 40 MM</td><td>Refer Section 2.0</td></tr><tr><td>ON LINE >= 50 MM</td><td>Refer Section 2.0</td></tr><tr><td>TEMP. CONNECTION</td><td>40 MM & BELOW</td><td>FLANGED</td></tr><tr><td>PRESS. CONNECTION</td><td>20 MM</td><td>SOCKET WELD WITH VALVE</td></tr></tbody></table> ElbowASME B 16.9, B 16.11 Fittings (Tees,Reducers, etc.)ASME B 16.9, B 16.11 Flanges / Blind FlangeANSI B 16.5 BoltingBolts: A193GR.B7, Nuts: A194 GR.2H GasketASME B 16.21								ITEMS	SIZE	DESCRIPTION	MAINTANANCE JOINTS	ALL	FLANGED, TO BE KEPT MINIMUM	PIPE JOINTS	40 MM & BELOW	SOCKET WELD COUPLING	50 MM & ABOVE	BUTT WELDED	DRAINS	ON LINE <= 40 MM	Refer Section 2.0	ON LINE >= 50 MM	Refer Section 2.0	VENTS	ON LINE <= 40 MM	Refer Section 2.0	ON LINE >= 50 MM	Refer Section 2.0	TEMP. CONNECTION	40 MM & BELOW	FLANGED	PRESS. CONNECTION	20 MM	SOCKET WELD WITH VALVE
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	METALTEC CONSULTANCY & SERVICES 202A,Earth-The Landmark Sun Pharma Road,Atriadra,Vadodara.india	PIPING MATERIAL SPECIFICATION									
		PIPE CLASS :N4S1									
						0	18.06.2015	SSH	SK	-	
DOC. NO : MCS/PMS/05-06 SHEET : 2 OF 2				REV.	DATE	PPD BY	CHKD BY	REMARKS			
Item	Description	Diameter Range		SCH/ Thk (mm) / Rating	Dimensional Standard	Material	Remarks				
		From	To								
PIPES											
	SEAMLESS	15	40	Sch. 40	ASME B 36.19	ASTM A312 TP 304L	PLAIN END				
	SEAMLESS	50	150	Sch. 10	ASME B 36.19	ASTM A312 TP 304L	BEVELLED END				
	E.F.S.W	200	200	4	ASME B 36.19	ASTM A312 TP 304L CL1	BEVELLED END				
	E.F.S.W	250	400	5	ASME B 36.19	ASTM A312 TP 304L CL1	BEVELLED END				
FLANGE											
	Weld Neck Flange	15	40	150	ASME B 16.5	ASTM A182 GR.F304L	RAISED FACE				
	LAP JOINT	50	400	150	ASME B 16.5	ASTM A 105	RAISED FACE				
	Blind	15	400	150	ASME B 16.5	ASTM A182 GR.F304L	RAISED FACE				
FITTINGS											
ELBOW											
	Elb. 45° & 90°	15	40	3000	ASME B 16.11	ASTM A 182 GR.F304L	SOCKET WELD				
	Elb. 45° & 90°	50	150	M	ASME B 16.9	ASTM A403 GR.WP304L-S	BEVELLED END				
	Elb. 45° & 90°	200	400	M	ASME B 16.9	ASTM A403 GR.WP304L WX	BEVELLED END				
TEE											
	EQUAL & REDUCING	15	40	3000	ASME B 16.11	ASTM A 182 GR.F304L	SOCKET WELD				
	EQUAL & REDUCING	50	150	M	ASME B 16.9	ASTM A403 GR.WP304L-S	BEVELLED END				
	EQUAL & REDUCING	200	400	M	ASME B 16.9	ASTM A403 GR.WP304L WX	BEVELLED END				
REDUCER											
	CONCENTRIC & ECCENTRIC	15	40	3000	ASME B 16.11	ASTM A 182 GR.F304L	PLAIN END				
	CONCENTRIC & ECCENTRIC	40	150	M	ASME B 16.9	ASTM A403 GR.WP304L-S	BEVELLED END				
	CONCENTRIC & ECCENTRIC	200	400	M	ASME B 16.9	ASTM A403 GR.WP304L WX	BEVELLED END				
COUPLING - FULL & HALF											
	COUPLING	15	40	3000	ASME B 16.11	ASTM A 182 GR.F304L	SOCKET WELD				
VALVE											
	GATE VALVE	15	40	800	API-602	BODY-ASTM A 182GRF304L, TRDM-SS304L	SOCKET WELD				
	GATE VALVE	50	400	150	API-600	BODY-ASTM A 315GRCF3 , TRDM - SS304L	RAISED FACE				
	GLOBE VALVE	15	40	800	BS-5332	BODY-ASTM A 182GRF304L, TRDM-SS304L	SOCKET WELD				
	GLOBE VALVE	50	400	150	BS-1873	BODY-ASTM A 315GRCF3 , TRDM - SS304L	RAISED FACE				
	CHECK VALVE	15	40	800	BS-5332	BODY-ASTM A 182GRF304L, TRDM-SS304L	SOCKET WELD				
	CHECK VALVE	50	400	150	BS-1868	BODY-ASTM A 315GRCF3 , TRDM - SS304L	RAISED FACE				
NUTS & BOLTS											
	BOLTS/NUTS	-	-	-	ASME B 18.2/B 18.22	BOLT-A193 GR. B7, NUT-A194 GR. 2H					
GASKETS											
	GASKET	15	400	4.4 MM	API -601	SS304L, SPR.WND+CAF					

	METALTEC	PIPING MATERIAL SPECIFICATION																																																																																	
	CONSULTANCY & SERVICES	PIPE CLASS :NBC1																																																																																	
	202A,Earth-The Landmark																																																																																		
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DOC. NO : MCS/PMS/06-06			0	18.06.2015	SSH	SK	-																																																																												
SHEET : 1 OF 2			REV.	DATE	PPD.BY	CHKD.BY	REMARKS																																																																												
PIPE CLASS : NBC1 FLANGE RATINGS : ASA 150 # BASE MATERIAL : CARBON STEEL CORROSION ALLOWANCE : 3.0 MM TEMPERATURE : 65°C Max. PRESSURE : 15 kg/cm² g (Max.) SERVICES : SULPHURIC ACID-(80-98.5% CONC.) NOTES : FLANGES ARE AS PER ANSI B 16.5, ASA 150 #, RAISED FACE. NUT BOLTS ARE STANDARD METRIC THREADED HOT DIP GALVANIZED AS PER IS 1367 Cl. 4 6/4 ALL DRAINS AND VENTS ARE AS PER SECTION 2.0 ALL INSTRUMENTS CONNECTIONS ARE AS PER P & ID's AND SHOULD BE FLANGE ENDS. 'M' DENOTES MATCH TO PIPE THICKNESS SPECIAL NOTES :<table><thead><tr><th>ITEMS</th><th>SIZE</th><th>DESCRIPTION</th></tr></thead><tbody><tr><td>MAINTANANCE JOINTS</td><td>ALL</td><td>FLANGED, TO BE KEPT MINIMUM</td></tr><tr><td>PIPE JOINTS</td><td>40 MM & BELOW</td><td>SOCKET WELD COUPLING</td></tr><tr><td></td><td>50 MM & ABOVE</td><td>BUTT WELDED</td></tr><tr><td>DRAINS</td><td>ON LINE <= 40 MM</td><td>Refer Section 2.0</td></tr><tr><td></td><td>ON LINE >= 50 MM</td><td>Refer Section 2.0</td></tr><tr><td>VENTS</td><td>ON LINE <= 40 MM</td><td>Refer Section 2.0</td></tr><tr><td></td><td>ON LINE >= 50 MM</td><td>Refer Section 2.0</td></tr><tr><td>TEMP. CONNECTION</td><td>40 MM & BELOW</td><td>FLANGED</td></tr><tr><td>PRESS. CONNECTION</td><td>20 MM</td><td>SOCKET WELD WITH VALVE</td></tr></tbody></table> <tr><td colspan="2">Elbow</td><td colspan="7">ASME B 16.9, B 16.11</td></tr> <tr><td colspan="2">Fittings (Tees,Reducers, etc.)</td><td colspan="7">ASME B 16.9, B 16.11</td></tr> <tr><td colspan="2">Flanges / Blind Flange</td><td colspan="7">ANSI B 16.5</td></tr> <tr><td colspan="2">Bolting</td><td colspan="7">ASME B 18.2/B 18.22</td></tr> <tr><td colspan="2">Gasket</td><td colspan="7">ASME B 16.21</td></tr>									ITEMS	SIZE	DESCRIPTION	MAINTANANCE JOINTS	ALL	FLANGED, TO BE KEPT MINIMUM	PIPE JOINTS	40 MM & BELOW	SOCKET WELD COUPLING		50 MM & ABOVE	BUTT WELDED	DRAINS	ON LINE <= 40 MM	Refer Section 2.0		ON LINE >= 50 MM	Refer Section 2.0	VENTS	ON LINE <= 40 MM	Refer Section 2.0		ON LINE >= 50 MM	Refer Section 2.0	TEMP. CONNECTION	40 MM & BELOW	FLANGED	PRESS. CONNECTION	20 MM	SOCKET WELD WITH VALVE	Elbow		ASME B 16.9, B 16.11							Fittings (Tees,Reducers, etc.)		ASME B 16.9, B 16.11							Flanges / Blind Flange		ANSI B 16.5							Bolting		ASME B 18.2/B 18.22							Gasket		ASME B 16.21						
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	METALTEC CONSULTANCY & SERVICES 202A, Earth-The Landmark Sun Pharma Road, Atladra, Vadodara, India	PIPING MATERIAL SPECIFICATION								
		PIPE CLASS :NBC1								
		DOC. NO : MCS/PMS/06-06 SHEET : 2 OF 2	0	18.06.2015	SSH	SK	-			
	REV.	DATE	PPD BY	CHKD BY	REMARKS					
Item	Description	Diameter Range		SCH/ Thk.(mm) / Rating	Dimensional Standard	Material	Remarks			
		From	To							
PIPES										
	SEAMLESS	15	40	Sch. 80	ASME B 36.10	ASTM A 106 Gr. B	PLAIN END			
	SEAMLESS	50	50	Sch. 80	ASME B 36.10	ASTM A 106 Gr. B	BEVELLED END			
	SEAMLESS	65	150	Sch. 40	ASME B 36.10	ASTM A 106 Gr. B	BEVELLED END			
FLANGE										
	SOCKET WELD FLANGE	15	40	150	ASME B 16.5	ASTM A 105	RAISED FACE			
	WELD NECK FLANGE	50	150	150	ASME B 16.5	ASTM A 105	RAISED FACE			
	BLDND	15	150	150	ASME B 16.5	ASTM A 105	RAISED FACE			
FITTINGS										
ELBOW										
	ELBOW 90°	15	40	3000	ASME B 16.11	ASTM A 105	SOCKET WELD			
	ELBOW 45°	15	40	3000	ASME B 16.11	ASTM A 105	SOCKET WELD			
	ELBOW 90°	50	150	M	ASME B 16.9	ASTM A 234 GR. WPB	BEVELLED END			
	ELBOW 45°	50	150	M	ASME B 16.9	ASTM A 234 GR. WPB	BEVELLED END			
TEE										
	EQUAL & REDUCING	15	40	3000	ASME B16.11	ASTM A 105	SOCKET WELD			
	EQUAL & REDUCING	50	150	M	ASME B16.9	ASTM A 234 GR. WPB	BEVELLED END			
REDUCER										
	CONCENTRIC	15	150	M	ASME B 16.9	ASTM A 234 GR. WPB	BEVELLED END			
	ECCENTRIC	15	150	M	ASME B 16.9	ASTM A 234 GR. WPB	BEVELLED END			
	SWAGE REDUCER	15	40	3000	ASME B 16.11	ASTM A 105	PLAIN END			
END CAP										
	SCREWED NPT	15	40	3000	ASME B 16.11	ASTM A 105				
	CAP	50	150	M	ASME B 16.9	ASTM A 234 GR. WPB,SCH40	BEVELLED END			
COUPLING - FULL & HALF										
	COUPLING	15	40	3000	ASME B 16.11	ASTM A 105	SOCKET WELD			
VALVE										
	GATE VALVE	15	40	800	API-602	BODY-ASTM A 105, TRIM-SS316	SOCKET WELD			
	GATE VALVE	50	300	150	API-600	BODY-ASTM A 216 GR.WCB., TRIM-SS316	FLANGED END , B-16.5,RAISED FACE			
	GLOBE VALVE	15	40	800	BS-5352	BODY-ASTM A 105, TRIM-SS316	SOCKET WELD			
	GLOBE VALVE	50	300	150	BS-1873	BODY-ASTM A 216 GR.WCB., TRIM-SS316	FLANGED END , B-16.5,RAISED FACE			
	CHECK VALVE	15	40	800	BS-5352	BODY-ASTM A 105, TRIM - SS316	SOCKET WELD			
	CHECK VALVE	50	300	150	BS-1868	BODY-ASTM A 216 GR.WCB., TRIM-SS316	FLANGED END, RAISED FACE			
	BALL VALVE	15	300	150	BS-5351	BODY-ASTM A216 GRWCB , SS316	RAISED FACE			
	PLUG VALVE	15	300	150	BS-5353	BODY- A 216 GR.WCB., TRIM-PLUG-SS316	FLANGED END, 150, ASME B-16.5			
NUTS & BOLTS										
	BOLTS/NUTS				ASME B 18.2/B 18.22	BOLT-A193 GR. B7, NUT-A194 GR. 2H				
GASKETS										
	GASKET	1/2"	14"	3 MM	ASME B 16.21	IS-2712-GR.A/1				
TRAP/STRAINER										
	PERMANENT STRAINER	15	40	800	-	BODY-ASTM A 105, INTERNAL-SS 316	SOCKET WELD, Y-TYPE,800			
	PERMANENT STRAINER	50	300	150	-	BODY-ASTM A 216 GR.WCB., INTERNAL - SS 316	BEVELLED END, T-TYPE			